

MACHINE LEARNING FROM A STATISTICAL PERSPECTIVE

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July 15, 2026

Can you identify the missing number?

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4	6	5	
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4	6	5	□
3	5	4	17
5	7	3	16

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5	7	3	16
4	6	6	32
7	9	5	38
1	4	7	27

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We **learn** the **model** (relationship) $Y = X_2 X_3 - X_1$ from the training data (black part) to predict the **output** Y (here the predicted value is 26) for the future **input** (X_1, X_2, X_3).

Can you identify the missing number?

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Y : Response variable, Output variable, Dependent variable.

X_1, X_2, X_3 : Predictor variables, Input variables, Independent variables, Covariates, Features, Measurement variables.

Can you identify the missing label (colour)?

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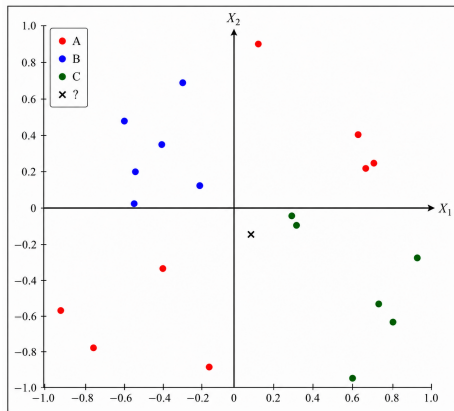
X_1	X_2	Y
0.28	-0.05	C
-0.45	-0.55	A
-0.47	0.35	B
0.78	-0.83	C
-0.16	-0.94	A
-0.56	0.01	B
-0.95	-0.60	A
0.30	-0.09	C
-0.56	0.18	B
0.62	-0.99	C
0.61	0.40	A
-0.32	0.69	B
0.91	-0.33	C
-0.81	-0.81	A
0.69	0.21	A
-0.61	0.46	B
0.07	0.95	A
-0.24	0.10	B
0.66	0.24	A
0.72	-0.75	C
0.07	-0.13	☐

Can you identify the missing label (colour)?

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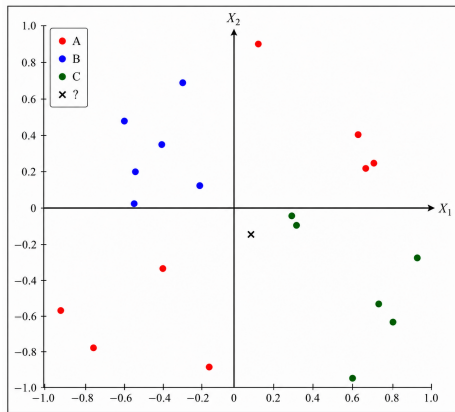


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Decision rule:

$X_1 X_2 > 0$	A
$X_1 < 0, X_2 > 0$	B
$X_1 > 0, X_2 < 0$	C

Can you identify the person??

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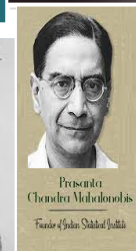
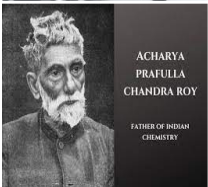
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Can you identify the person??

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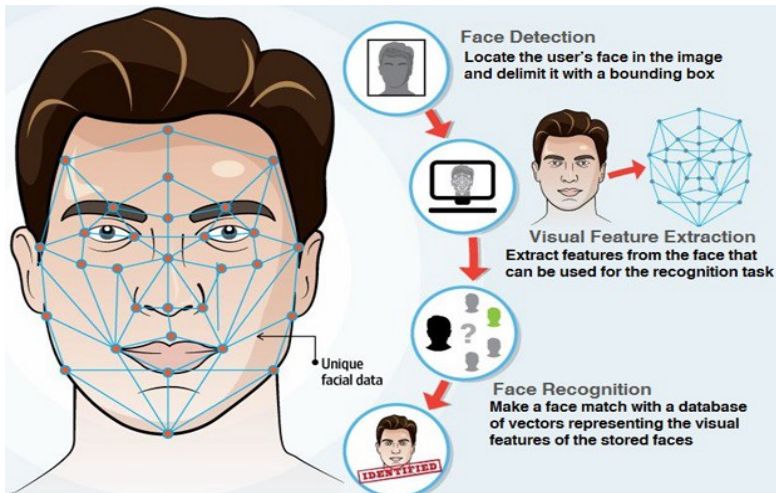
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Face recognition: How does the machine work?

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Some examples of human learning

- A newborn learns the voices and faces of the parents.
- A toddler learns shapes, colours, digits and the alphabet.
- We decide whether to cross the road.
- We decide when to start from home to catch the train.
- Medical practitioners analyse previous data and read medical journals to learn more about a new disease.

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Learning: Study of 'input-output' relationship

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Learning: Study of 'input-output' relationship

Machine Learning: Development of algorithms so that machines can learn.

Statistical Machine Learning: Development of the algorithms based on statistical theories.

Different types of learning

- Supervised (Training data contains information about both input and output (target) variables)
 - Regression (Response is numerical)
 - Classification (Response is categorical)

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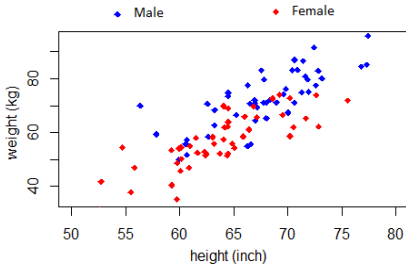
Different types of learning

- Supervised (Training data contains information about both input and output (target) variables)
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- Unsupervised
 - Cluster Analysis (Unsupervised Classification)
 - Principal Component Analysis
 - Multidimensional Scaling and Self-organizing Maps
 - Independent Component Analysis

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- Semi-supervised learning
- Reinforcement learning
- Transfer learning

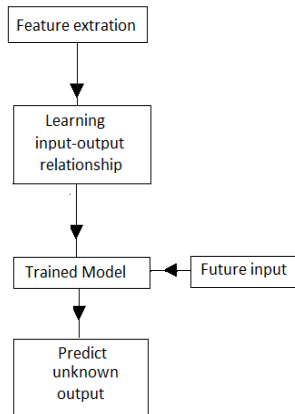
Supervised Learning: Regression and Classification

Machine learning aims at developing algorithms for more accurate predictions based on the data.



Gender	Height	Weight
Male	69	
Female	61	
	63	60
	70	74

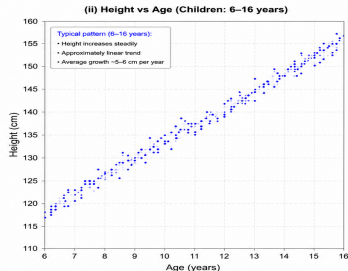
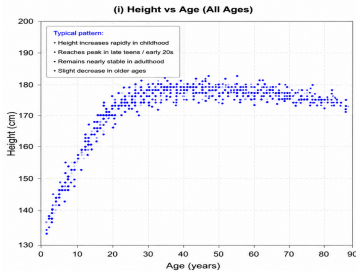
	Regression
	Response is numerical
	Classification
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Regression

- Y: Response variable, $X_1, X_2, \dots, X_d$: Covariates

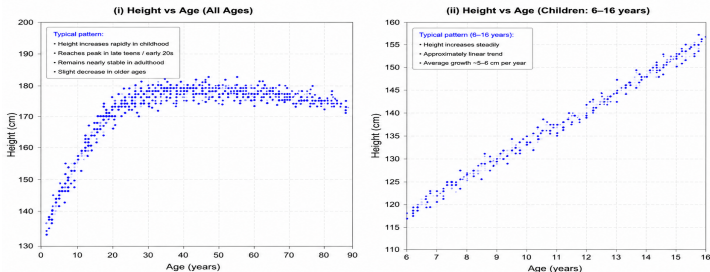
Height vs Age: Two Different Age Ranges



Regression

- Y: Response variable, $X_1, X_2, \dots, X_d$: Covariates

Height vs Age: Two Different Age Ranges



- We assume the model

$$Y = \underbrace{f(X_1, X_2, \dots, X_d)}_{\text{signal}} + \underbrace{\epsilon}_{\text{noise}}$$

and look for estimating the function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ from the data.

- Both parametric and nonparametric methods can be used for the estimation of the regression function/surface f .

Estimation of f : minimisation of expected loss

- We can define a suitable loss function $L(a, b)$ and look for minimising the expected loss $E[L(Y, \hat{f}(X_1, X_2, \dots, X_d))]$.

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Estimation of f : minimisation of expected loss

- We can define a suitable loss function $L(a, b)$ and look for minimising the expected loss $E[L(Y, \hat{f}(X_1, X_2, \dots, X_d))]$.
- **Squared error loss: $L(a, b) = (a - b)^2$**
 - Risk($\hat{f}$) = $E[(Y - \hat{f}(X_1, X_2, \dots, X_d))^2]$ is minimised when $\hat{f}(x_1, x_2, \dots, x_d) = E(Y \mid X_1 = x_1, X_2 = x_2, \dots, X_d = x_d)$
 - Risk($\hat{f}$) = $E[(Y - \hat{E}[\hat{f}(X_1, \dots, X_d)])^2] + \text{Var}[\hat{f}(X_1, \dots, X_d)]$
= Squared bias + Variance.

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= Squared bias + Variance.
- **Absolute error loss: $L(a, b) = |a - b|$**
 - Risk($\hat{f}$) = $E|(Y - \hat{f}(X_1, X_2, \dots, X_d))|$ is minimised when $\hat{f}(x_1, x_2, \dots, x_d) = \text{Med}(Y \mid X_1 = x_1, \dots, X_d = x_d)$
 - Risk($\hat{f}$) $\leq E|Y - \text{Med}[\hat{f}(X_1, \dots, X_d)]| + MD[\hat{f}(X_1, \dots, X_d)]$
= Absolute bias + Mean Deviation.

Estimation of f : minimisation of expected loss

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= Absolute bias + Mean Deviation.
- **In practice, we minimise the empirical version of expected loss**

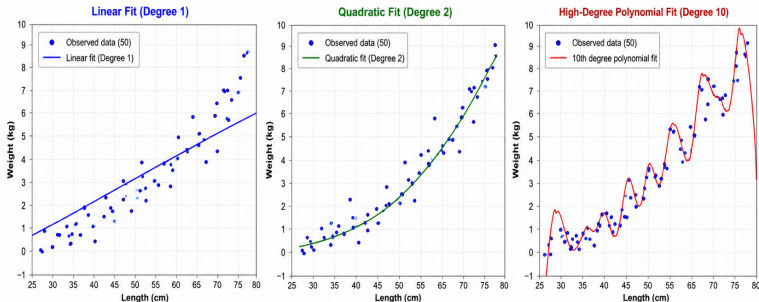
$$\frac{1}{n} \sum_{\ell=1}^n L(y_{\ell}, \hat{f}(X_{1\ell}, X_{2\ell}, \dots, X_{d\ell})) \text{ or } \sum_{\ell=1}^n L(y_{\ell}, \hat{f}(x_{1\ell}, x_{2\ell}, \dots, x_{d\ell}))$$

Bias-variance tradeoff: Underfit and overfit

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Salmon Fish Length vs Weight (50 Observations)
Least Squares Fits: Linear (Degree 1) vs Quadratic (Degree 2) vs High-Degree Polynomial (Degree 10)



Underfitting (High Bias)

The linear model cannot capture the curvature in the data.
Systematic pattern remains in residuals.

Training Performance (Least Squares)

SSE (Sum of Squared Errors) = 69.72
 R^2 (Coefficient of Determination) = 0.646
RMSE (Root MSE) = 1.180

Good Fit (Balanced Bias-Variance)

Captures the nonlinear trend well without chasing the random noise.
Best generalization.

Training Performance (Least Squares)

SSE (Sum of Squared Errors) = 24.18
 R^2 (Coefficient of Determination) = 0.880
RMSE (Root MSE) = 0.694

Overfitting (High Variance)

Follows the noise and random fluctuations (especially at the ends).
Poor generalization to new data.

Training Performance (Least Squares)

SSE (Sum of Squared Errors) = 3.89
 R^2 (Coefficient of Determination) = 0.980
RMSE (Root MSE) = 0.278

Bias-Variance Trade-off

As model complexity increases: Bias decreases but Variance increases.
The quadratic model provides the best trade-off on this dataset.

Linear regression

We assume the model $f(x_1, \dots, x_d) = \beta_0 + \beta_1 x_1 + \dots + \beta_d x_d$.

Data matrix:

	X_1	X_2	$\dots$	X_d	Y
1	x_{11}	x_{21}	$\dots$	x_{d1}	y_1
2	x_{12}	x_{22}	$\dots$	x_{d2}	y_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
ℓ	$x_{1\ell}$	$x_{2\ell}$	$\dots$	$x_{d\ell}$	y_ℓ
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
n	x_{1n}	x_{2n}	$\dots$	x_{dn}	y_n

We minimize

$$\sum_{\ell=1}^n (y_\ell - \beta_0 - \beta_1 x_{1\ell} - \dots - \beta_d x_{d\ell})^2 : \text{least square (LS) regression}$$

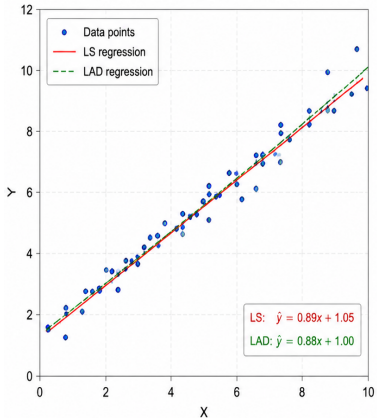
$$\sum_{\ell=1}^n |y_\ell - \beta_0 - \beta_1 x_{1\ell} - \dots - \beta_d x_{d\ell}| : \text{least absolute dev. (LAD) regression}$$

LS and LAD regression

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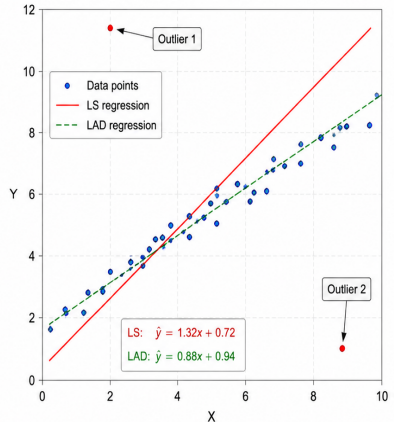
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(i) Normal Data (No Outliers)



Both LS and LAD give similar fit when there are no outliers.

(ii) Contaminated Data (With Outliers)



LS is strongly affected by outliers, while LAD is much more robust.

LAD regression: Estimation of parameters

Using the following facts, one can develop an iterative algorithm to minimise $\sum_{\ell=1}^n |y_{\ell} - \beta_0 - \beta_1 x_{1\ell} - \dots - \beta_d x_{d\ell}|$

- Fact 1: $\sum_{\ell=1}^n |z_{\ell} - \alpha_0|$ is minimised when α_0 is the median of $\{z_1, z_2, \dots, z_n\}$.
- Fact 2: $\sum_{\ell=1}^n |z_{\ell} - \alpha w_{\ell}| = \sum_{\ell=1}^n |w_{\ell}| |(z_{\ell}/w_{\ell}) - \alpha|$ is minimized when α is the median of the following frequency distribution

Value	z_1/w_1	z_2/w_2	$\dots$	z_{ℓ}/w_{ℓ}	$\dots$	z_n/w_n
Frequency	$ w_1 $	$ w_2 $	$\dots$	$ w_{\ell} $	$\dots$	$ w_n $

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Another option: **iteratively re-weighted least squares (IRLS)** algorithm, where one starts with $\beta^{(1)} = (\beta_0^{(1)}, \beta_1^{(1)}, \dots, \beta_d^{(1)})$ and at the $t+1$ -th ($t \geq 1$) step, gets $\beta^{(t+1)}$ by minimising

$$\psi_t(\beta) = \sum_{\ell=1}^n w_{\ell}^{(t)} (y_{\ell} - \mathbf{x}_{\ell}^{\top} \beta)^2, \text{ where } w_{\ell}^{(t)} = \left| y_{\ell} - \mathbf{x}_{\ell}^{\top} \beta^{(t)} \right|^{-1}.$$

and $\mathbf{x}_{\ell} = (1, x_{1\ell}, x_{1\ell}, \dots, x_{d\ell})$.

LS Regression: Estimation of Parameters

Define

$$\mathbb{X} = \begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{d1} \\ 1 & x_{12} & x_{22} & \cdots & x_{d2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1\ell} & x_{2\ell} & \cdots & x_{d\ell} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{dn} \end{bmatrix}, \quad \mathbb{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_\ell \\ \vdots \\ y_n \end{bmatrix} \quad \text{and} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \vdots \\ \beta_d \end{bmatrix}.$$

$$\text{Minimizing } \sum_{\ell=1}^n (y_\ell - \beta_0 + \beta_1 x_{1\ell} + \cdots + \beta_d x_{d\ell})^2 = \|\mathbb{Y} - \mathbb{X}\beta\|^2,$$

one gets the least square estimate $\hat{\beta}_{LS} = (\mathbb{X}^\top \mathbb{X})^{-1} \mathbb{X}^\top \mathbb{Y}$.

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Under standard model assumptions, we have

- $\text{Disp}(\hat{\beta}_{LS}) = \sigma^2 (\mathbb{X}^\top \mathbb{X})^{-1}$.
- $E\|\hat{\beta}_{LS} - \beta\|^2 = \sum_{k=0}^d \text{Var}(\hat{\beta}_k) = \sigma^2 \sum_k (1/\lambda_k)$, where the λ_k 's are the eigenvalues of $\mathbb{X}^\top \mathbb{X}$

The problem of multicollinearity: An example

Consider a problem with two covariates where X_1 , X_2 and Y are standardised (each of them has mean 0 and variance 1)

- No need to consider the intercept term β_0 here and one can consider the linear model $Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon$.
- Here $\mathbb{X}^T \mathbb{X} = n \begin{bmatrix} 1 & r \\ r & 1 \end{bmatrix}$ and $\mathbb{X}^T \mathbb{Y} = n \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$, where $r = \text{corr}(X_1, X_2)$, $r_1 = \text{corr}(Y, X_1)$ and $r_2 = \text{corr}(Y, X_2)$.
- $\hat{\beta}_{LS} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \frac{1}{1-r^2} \begin{bmatrix} r_1 - r r_2 \\ r_2 - r r_1 \end{bmatrix}$.
- $\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \sigma^2 / [n(1 - r^2)]$.

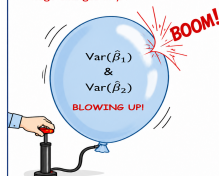
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- $\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \sigma^2 / [n(1 - r^2)]$.
- If $r \approx \pm 1$,
 - 1 $|\hat{\beta}_1|$ and $|\hat{\beta}_2|$ become large.
 - 2 Signs of $\hat{\beta}_1$ and $\hat{\beta}_2$ can be misleading.
 - 3 These estimates become less reliable (high variance).

• With two nearly collinear covariates, the design matrix $X^T X$ is nearly singular.

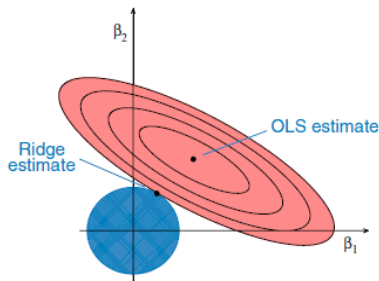
Small changes in the data cause huge changes in $\hat{\beta}$.



As collinearity increases ($r \rightarrow \pm 1$), the variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ explode!

Ridge regression

We minimize $\|\mathbb{Y} - \mathbb{X}\beta\|^2$ subject to $\|\beta\|^2 \leq C$ for some $C > 0$.

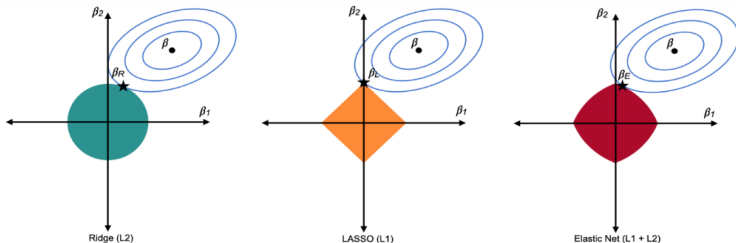


$$\|\mathbb{Y} - \mathbb{X}\beta\|^2 = \|\mathbb{Y} - \mathbb{X}\hat{\beta}_{LS}\|^2 + (\beta - \hat{\beta}_{LS})^\top \mathbb{X}^\top \mathbb{X} (\beta - \hat{\beta}_{LS}).$$

- This minimisation is equivalent to the minimisation of $\|\mathbb{Y} - \mathbb{X}\beta\|^2 + \lambda(\|\beta\|^2 - C)$ for some $\lambda > 0$.
- Ridge solution: $\hat{\beta}_R = (\mathbb{X}^\top \mathbb{X} + \lambda \mathbf{I})^{-1} \mathbb{X}^\top \mathbb{Y}$.

LASSO regression

We minimize $\|Y - X\beta\|^2$ subject to $\|\beta\|_1 \leq C$ for some $C > 0$, or equivalently, minimize $\|Y - X\beta\|^2 + \lambda(\|\beta\|_1 - C)$ for some $\lambda > 0$.

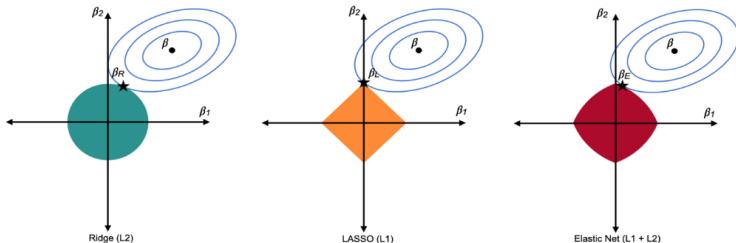


Advantage: Unlike ridge regression, it automatically deletes some of the unimportant variables from the model.

Disadvantage: No closed-form expression. Usually, the coordinate-descent algorithm is used for minimisation.

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Elastic net: considers a combination of ℓ_1 and ℓ_2 penalties. It minimizes $\|\mathbb{Y} - \mathbb{X}\beta\|^2 + \lambda(\frac{\alpha}{2}\|\beta\|^2 + (1 - \alpha)\|\beta\|_1)$, $\lambda > 0, \alpha \in [0, 1]$.

Nonparametric regression

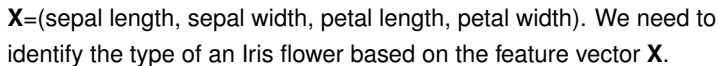
- Piecewise constant function (naive method)
- Piecewise linear/quadratic/cubic functions (splines)
- Additive models
- Projection pursuit regression
- Artificial Neural Networks (Perceptron models)
- Generalised additive models
- Regression trees
- MARS (Multivariate Adaptive Regression Splines)
- Nearest neighbour regression
- LOESS regression
- Kernel regression
- Local linear and local polynomial regression
- Support vector regression
- Wavelets regression

Classification problem: An example

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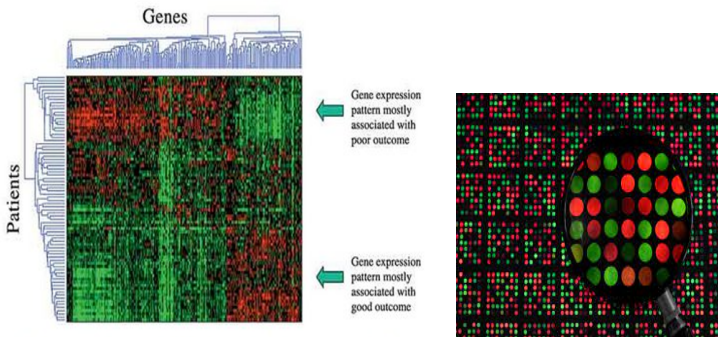
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Classification problem: Another example

- Breast cancer data: For each patient (normal patient and cancer patients), expression levels are observed for 6137 genes



$\mathbf{X}$ = microarray expression levels of 6137 genes.

We need to classify a new patient based on the feature vector $\mathbf{X}$.

Classification: mathematical framework

Data: **Class-1:** $\mathbf{X}_{11}, \mathbf{X}_{12}, \dots, \mathbf{X}_{1n_1}$

Class-2: $\mathbf{X}_{21}, \mathbf{X}_{22}, \dots, \mathbf{X}_{2n_2}$

$\vdots$

Class- J : $\mathbf{X}_{J1}, \mathbf{X}_{J2}, \dots, \mathbf{X}_{Jn_J}$

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We want to construct a decision rule $\delta : \mathbb{R}^d \rightarrow \{\mathbf{1}, \mathbf{2}, \dots, \mathbf{J}\}$
i.e., given any $\mathbf{x} \in \mathbb{R}^d$, we want to find its label.

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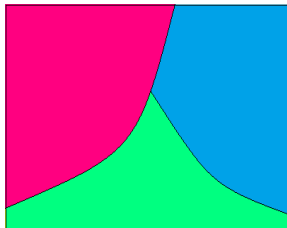
Class-2: $\mathbf{X}_{21}, \mathbf{X}_{22}, \dots, \mathbf{X}_{2n_2}$

$\vdots$

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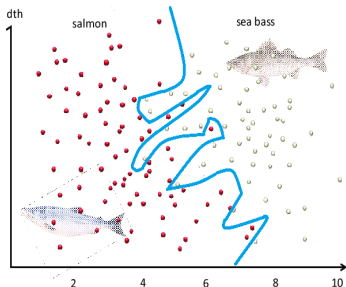
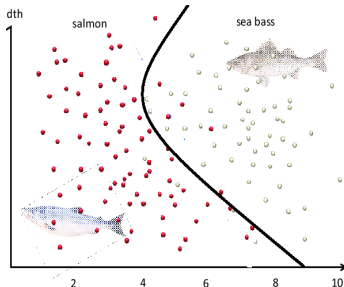
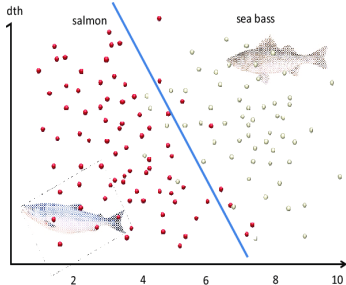
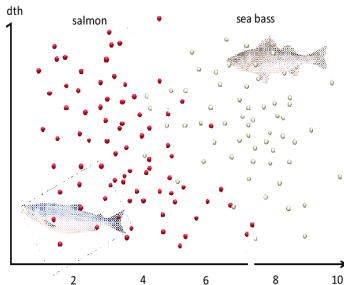
A decision rule forms a finite partition of the feature space.



Linear and nonlinear classifiers

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Loss function in classification

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Consider a J -class problem,

- The response variable Y takes values in the set $\{1, 2, \dots, J\}$.
- Decision rule: $\delta(X_1, X_2, \dots, X_d) : \mathbb{R}^d \rightarrow \{1, 2, \dots, J\}$.

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 $L(\delta(X_1, X_2, \dots, X_d), Y) = \mathbb{I}\{\delta(X_1, X_2, \dots, X_d) \neq Y\}$, the 0 – 1
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- Expected loss: $\text{Risk}(\delta) = E[L(\delta(X_1, X_2, \dots, X_d), Y)]$
 $= P(\delta(X_1, X_2, \dots, X_d) \neq Y)$
- Empirical version: $\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{\delta(x_{1i}, x_{2i}, \dots, x_{di}) \neq y_i\}$, the proportion of misclassification in the training set, is not a convex function.

How to form the partition?: An illustrative example (Example 1)

In a bolt factory, **Machine-1** and **Machine-2** manufacture **40%** and **60%** of total bolts. Of their outputs, **20%** and **10%**, respectively, are defective bolts. In a pack of 20 bolts manufactured by the same machine, 3 are found to be defective. Which machine is more likely to manufacture the bolts in the pack?

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$$f_1(x) = \binom{20}{x} (0.2)^x (0.8)^{20-x} \text{ for } x = 0, 1, 2, \dots, 20.$$

- $X | M_2 \sim \text{Bin}(20, 0.1)$

$$f_2(x) = \binom{20}{x} (0.1)^x (0.9)^{20-x} \text{ for } x = 0, 1, 2, \dots, 20.$$

Example-1 (contd.)

- Bayes theorem:-

$$p(1 | x) = P(M_1 | X = x) = \frac{P(M_1)P(X=x|M_1)}{\sum_{i=1}^2 P(M_i)P(X=x|M_i)}$$

$$= \pi_1 f_1(x) / \sum_{i=1}^2 \pi_i f_i(x).$$

$$p(2 | x) = \pi_2 f_2(x) / \sum_{i=1}^2 \pi_i f_i(x).$$

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- Decision: Here $x = 3$, $p(1 | x) = 0.419$ and $p(2 | x) = 0.581$.
The bolts in the pack are more likely to be from Machine-2.

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x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\delta(x)$	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

The decision rule derived from the Bayes theorem (Bayes rule) δ forms a partition of the feature space.

{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}.

Example-2

A bulb factory manufactures two types of electric bulbs. The lifetime of a bulb follows an exponential distribution with mean 2000 and 3000 hours for Type-1 and Type-2, respectively. Two-third of the total production are of Type-1 and the rest are of Type-2. A person bought two bulbs of the same type and their lifetimes were recorded as 2400 hours and 3600 hours, respectively. Can you guess the type of the bulbs bought by the person?

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$$\pi_1 = P(T_1) = 2/3, \pi_2 = P(T_2) = 1/3.$$

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- $\mathbf{X} = (X_1, X_2)$: Feature vector (lifetimes of the two bulbs)
- Population distributions (probability density functions):-

- $X_1, X_2 \mid T_1 \stackrel{iid}{\sim} \exp(2000)$

$$f_1(\mathbf{x}) = f_1(x_1, x_2) = \left(\frac{1}{2000}\right)^2 e^{-(x_1+x_2)/2000} \text{ for } x_1, x_2 \geq 0.$$

- $X_1, X_2 \mid T_2 \stackrel{iid}{\sim} \exp(3000)$

$$f_2(\mathbf{x}) = f_2(x_1, x_2) = \left(\frac{1}{3000}\right)^2 e^{-(x_1+x_2)/3000} \text{ for } x_1, x_2 \geq 0.$$

Example-2 (contd.)

- Bayes theorem:-

$$p(1 | \mathbf{x}) = \pi_1 f_1(\mathbf{x}) / \sum_{i=1}^2 \pi_i f_i(\mathbf{x}) \quad p(2 | \mathbf{x}) = \pi_2 f_2(\mathbf{x}) / \sum_{i=1}^2 \pi_i f_i(\mathbf{x})$$

$$\begin{aligned}
 p(1 | \mathbf{x}) \geq p(2 | \mathbf{x}) &\Leftrightarrow \pi_1 f_1(\mathbf{x}) \geq \pi_2 f_2(\mathbf{x}) \Leftrightarrow \frac{\pi f_1(\mathbf{x})}{\pi_2 f_2(\mathbf{x})} \geq 1 \\
 &\Leftrightarrow 2\left(\frac{3}{2}\right)^2 e^{-(x_1+x_2)\left(\frac{1}{2000} - \frac{1}{3000}\right)} \geq 1 \Leftrightarrow x_1 + x_2 \leq 9024.46
 \end{aligned}$$

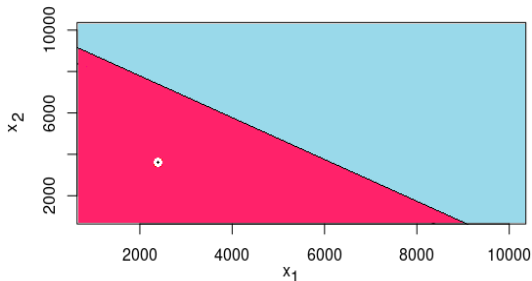
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- Decision: Here $\mathbf{x} = (x_1, x_2) = (2400, 3600)$, $x_1 + x_2 = 6000$. It is more likely that the bulbs are of **Type-1**.



Bayes rule forms a linear partition of the feature space.

How to find the best classifier (best partition) ?

Consider a two class problem.

π_1 and π_2 : prior probabilities.

$f_1(\cdot)$ and $f_2(\cdot)$: probability density/mass functions.

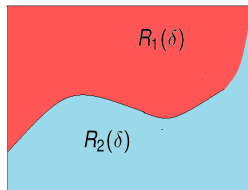
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	Class-1	Class-2
Class-1	OK	Error-2
Class-2	Error-1	OK



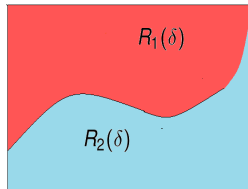
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Expected loss (misclassification probability) of the decision rule (classifier) δ is given by

$$\begin{aligned}
 \text{Risk}(\delta) &= \pi_1 \int_{R_2(\delta)} f_1(\mathbf{x}) d\mathbf{x} + \pi_2 \int_{R_1(\delta)} f_2(\mathbf{x}) d\mathbf{x} \\
 &= \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})] d\mathbf{x}
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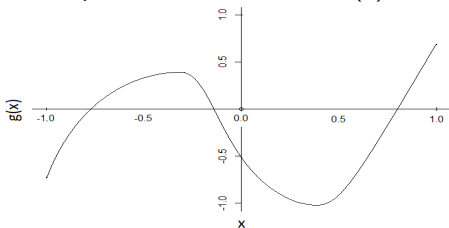
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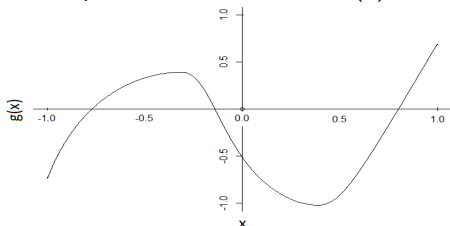
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We know that $\int_A g(\mathbf{x}) d\mathbf{x}$ is minimum when $A = \{\mathbf{x} : g(\mathbf{x}) < 0\}$.

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- $\mathcal{E}(\delta) = \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(\mathbf{x}) - \pi_1 f_1(\mathbf{x})] d\mathbf{x}$ will be minimum if
 $R_1(\delta) = \{\mathbf{x} : \pi_1 f_1(\mathbf{x}) > \pi_2 f_2(\mathbf{x})\}$, i.e., $\mathbf{x}$ is classified to Class-1
 if $\pi_1 f_1(\mathbf{x}) > \pi_2 f_2(\mathbf{x})$ or $p(1 | \mathbf{x}) > p(2 | \mathbf{x})$.
- The decision rule δ thus obtained is called the Bayes rule/
Bayes classifier/ Oracle Classifier.

Bayes classifier

- Notations:-

π_j : prior probability of the j -th class ($j = 1, 2, \dots, J$).

$f_j(\cdot)$: density/mass function of the j -th class.

$p(j | \mathbf{x})$: posterior probability of the j -th class given an observation $\mathbf{x}$.

- Bayes (Oracle) classifier :-

$$\delta(\mathbf{x}) = j \text{ if } p(j | \mathbf{x}) > p(i | \mathbf{x}) \text{ for all } i \neq j \text{ or}$$

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- Bayes rule can't be used in practice.
- We need to estimate the posterior probabilities or the density/mass functions from the data.

Some popular classification methods

- Classification using estimates of density functions
 - Parametric: linear and quadratic discriminant analysis.
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- Other classifiers
 - Support vector machine (SVM)

It assumes a specific parametric form of the separating surface and then optimizes a suitable objective function to estimate the associated parameters.

Classification under regression framework

Consider a two-class classification problem. Define $Y = 1$ if the observation comes from the first class, and 0 otherwise.

Note that we can classify $\mathbf{x}$ to class-1 if $E(Y | \mathbf{x}) = p(1 | \mathbf{x}) > 0.5$. So, one can regression Y on $\mathbf{X}$ to estimate $p(1 | \mathbf{x})$.

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Regression methods that can be used

- Kernel regression.
- Nearest neighbour regression.
- Artificial Neural Network (ANN).
- Regression tree

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Since linear regression does not guarantee that $\hat{p}(1 | \mathbf{x})$ lies between 0 and 1, it is not used for classification.

Regression methods that can be used

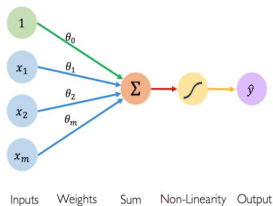
- Kernel regression.
- Nearest neighbour regression.
- Artificial Neural Network (ANN).
- Regression tree

For J -class problem, define J indicators, one for each class.

ANN: Single-layer and Multi-layer perceptrons

MACHINE
LEARNING

Anil K. Ghosh

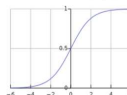


Activation Functions

$$\hat{y} = g(\theta_0 + \mathbf{X}^T \boldsymbol{\theta})$$

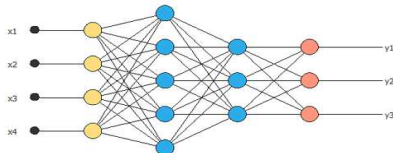
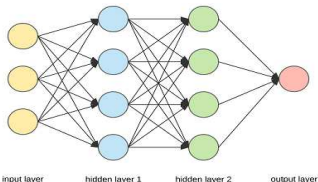
- Example: sigmoid function

$$g(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$$



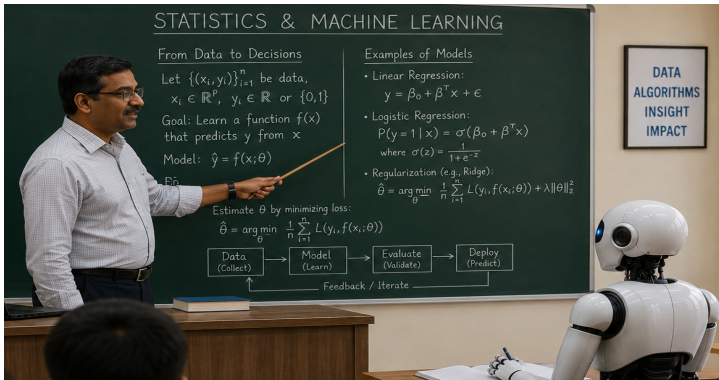
Single-layer perceptron model leads to linear classification.

For estimating more complex class boundaries, we use multi-layer perceptrons with one or more hidden layers.



References

- 1 Duda, R. O., Hart, P. E. and Stork, D. G. (2012). *Pattern Classification*. Wiley, New York.
- 2 Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*. Wiley, New York.



Statistical Learning

Lecture 02 Notes

1: Learning Definitions

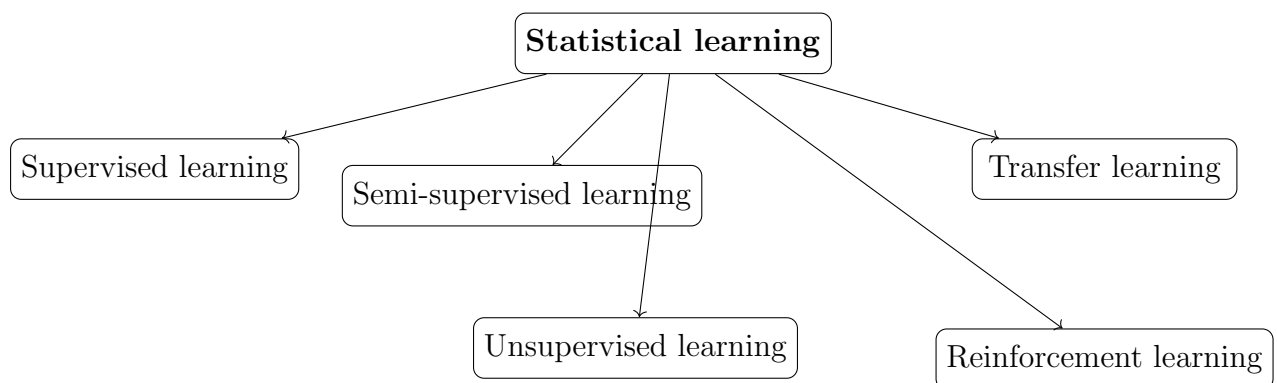
- **Learning** :- Extracting Knowledge / useful information from the data.
- **Mach. learning** :- We need to develop algorithms so that the machine Can do this job.
- **Statistical Mach. learning** :- We use statistical theories / methodologies to develop these algorithms.

2: What is Statistical Learning

Statistical Learning :-

It is branch of science where statistical theories and methodologies are used to develop algorithms for extracting meaningful information from the data.

3: Types of Statistical Learning (Flowchart)



4: Supervised Learning

Here we have a training set containing information on input (covariates) and output (response) variables and our aim is to study the relationship between the output and input variables so that we can predict the output for a new input as accurately as possible.

The output variable can be numerical or categorical.

- If we deal with a numerical response (output) it is called a **regression problem**.
- If the response is categorical, it is called a **classification problem**.

Formally, training set = $\{(\mathbf{X}_i, Y_i), i = 1, \dots, n\}$ where $\mathbf{X}_i \in \mathbb{R}^p$.

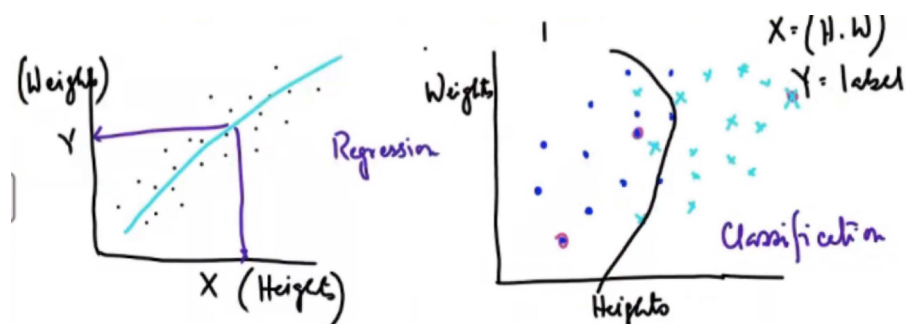
5: Unsupervised Learning

Here we do not have any labelled observation (response is not available) but still want to extract meaningful information from the data.

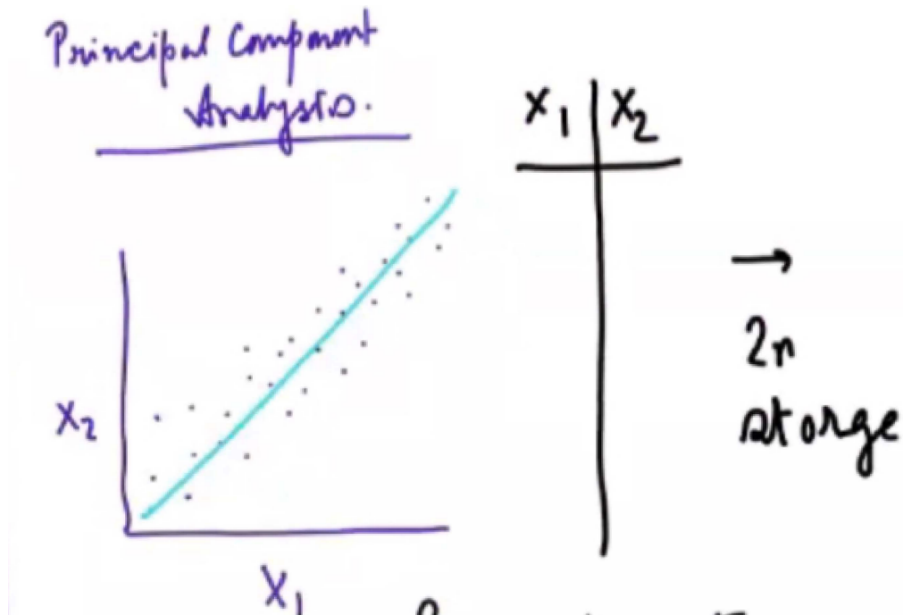
Common methods:

- Principal Component Analysis
- Multi dimensional Scaling
- Independent Component Analysis
- Cluster analysis

6: Regression vs Classification



7: Principal Component Analysis (PCA)



Assume $(0, 0)$ is the centre.

We look for direction vector $\alpha = (\alpha_1, \alpha_2)^T$ such that var of $\alpha^T \mathbf{x} = \alpha_1 X_1 + \alpha_2 X_2$ is maximized subject to

$$\|\alpha\| = \sqrt{\alpha_1^2 + \alpha_2^2} = 1.$$

Approximation:

$$\begin{pmatrix} x_{i1} \\ x_{i2} \end{pmatrix} \approx l_i \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

New storage: direction vector = 2, data points = $n \Rightarrow n + 2$

Also PCA can be viewed as finding lower dim representation that minimizes the error sum of squares (perpendicular residuals).

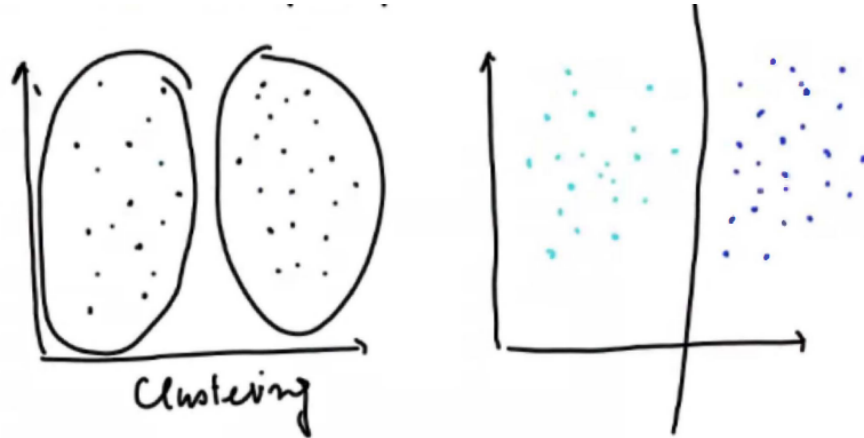
8: Other Methods

Multi-dimensional scaling :: It is also dimension reduction technique which considers linear/nonlinear projection of the observations such that the pairwise distances remains almost the same

Independent component Analysis (Blind source separation) :: It is method for extracting independent original signals from a data set containing mixtures of signals

9: Cluster Analysis

Cluster Analysis :: (unsupervised classification) Divide the data set into some homogeneous groups



Classification	Clustering
We have a training set of labelled observations	We do not have any labelled obsn.
Our aim to construct a discriminatory surface so that we classify the future (unlabelled) obsn as accurately as possible	Our aim is to divide this data set into several groups to understand the data better
The no. of classes are Known	No. of groups are unknown

10: Semi-supervised Learning



11: References and Evaluation

Ref:-

- Hastie, Tibshirani, Friedman : Elements of Stat. learning
- Duda, Hart, Stork : Pattern Classification.

Evaluation: 50 : Endsem, 30 : Midsem, 20 : Assign

SL-1, W02-L01

Learning paradigms and regression

Transfer learning: applying the knowledge obtained from one problem or domain to solve a new but related problem instead of starting from scratch.

Reinforcement learning: It is a learning process through trial and error, with rewards for good decisions and penalties for bad decisions.

Supervised learning:

- Regression
- Classification

Regression

Model:

$$Y = f(x) + \varepsilon \text{response} = \text{signal} + \text{noise}$$

In regression, our aim is to estimate the signal/regression function f .

In order to estimate f , we need to consider a loss function and minimize the expected loss.

Popular loss functions:

- Squared error loss

$$L(Y, f(x)) = (Y - f(x))^2$$

- Absolute error loss

$$L(Y, f(x)) = |Y - f(x)|$$

We should try to find that minimizes Risk $= R(f) = E(L(Y, f(x)))$, the expected loss. In practice, we minimize the empirical analog of $E(L(Y, f(x)))$. This is roughly

$$\frac{1}{n} \sum L(Y_i, f(x_i)),$$

where $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent observations on (X, Y) .

For example, in the case of squared error loss, we minimize

$$\frac{1}{n} \sum (Y_i - f(x_i))^2 = \sum (Y_i - f(x_i))^2.$$

Similarly, for absolute error loss function, we minimize

$$\sum |Y_i - f(x_i)|.$$

Bias-variance tradeoff and parametric regression

Model $Y = X + \varepsilon$ where $\varepsilon(t) \sim (0, \sigma^2)$

$$f(x) = ax + b \longrightarrow \text{underfit } \{\text{high bias, low var}\}$$

$$f(x) = ax^2 + bx + c \longrightarrow \text{fit}$$

$$f(x) = a_0x^0 + a_1x^1 + \dots + a_kx^k \longrightarrow \text{overfit } \{\text{high var, low bias}\}$$

$$E(Y - \hat{f}(x))^2 = E(Y - E(\hat{f}(x)))^2 + E(\hat{f}(x) - E(\hat{f}(x)))^2 = \text{bias}^2 + \text{variance}$$

f can be estimated either parametrically or nonparametrically.

In the case of parametric regression we assume a finite dimensional parametric form for f and estimate this parameters from the data.

For instance, in the case of linear regression, we assume f to be linear, i.e.

$$f(x) = \alpha + \beta x$$

and estimate this parameters by min

$$\sum (Y_i - \alpha - \beta x_i)^2$$

or

$$\sum |Y_i - \alpha - \beta x_i|.$$

We can think of even a simpler model $Y = \beta x$, where one minimizes

$$\sum (Y_i - \beta x_i)^2 \quad \text{or} \quad \sum |Y_i - \beta x_i|.$$

Simple model, LS weights, and LAD

$$\beta_1 = \frac{Y_1}{X_1}$$

$$\beta_2 = \frac{Y_2}{X_2}$$

$$\vdots$$

$$\beta_n = \frac{Y_n}{X_n}$$

$$\hat{\beta}_{LS} = \frac{\sum X_i Y_i}{\sum X_i^2} = \frac{\sum X_i^2 \beta_i}{\sum X_i^2} = \frac{\sum w_i \beta_i}{\sum w_i} \quad \text{where } w_i \propto X_i^2$$

$\hat{\beta}_{LAD}$ is the median of the following frequency distribution:

- Values $\beta_1, \beta_2, \dots, \beta_n$
- Weights $|X_1|, |X_2|, \dots, |X_n|$

$$|X_i| > \frac{1}{2} \sum |X_j|$$

$$\beta_1, \beta_2, \dots, \beta_n \quad |X_1|, \dots$$

Example:

$$X : 1, 2, 4, 8, \dots, 2^k$$

$$Y : 1, 2, 3, 4, \dots, k+1$$

$$\beta = Y/X : 1/1, 2/2, 3/4, 4/8, \dots, (k+1)/2^k$$

$$\beta_1, \beta_2, \beta_3, \dots, \beta_{k+1}$$

$$\min \sum |Y_i - \beta X_i|$$

$$\hat{\beta}_{LAD} = \frac{k+1}{2^k}$$

Another example:

$$X : 3, 4, 5, 7, 9$$

$$Y : 5, 10, 10, 30, 5$$

$$\text{Sum: } 6, 15, 25, 55, 60$$

For the general case, i.e. for the model

$$Y = \alpha + \beta X + \varepsilon$$

we have

$$\hat{\beta}_{LS} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\hat{\alpha}_{LS} = \bar{Y} - \hat{\beta}_{LS} \bar{X}$$

$(\hat{\alpha}_{LAD}, \hat{\beta}_{LAD})$ does not have a closed form expression.

$\sum |Y_i - \alpha - \beta X_i|$ is minimized using an iterative algo (namely the IRWLS algo is used).

Properties of LS solutions

Consider a model:

$$Y_i = \alpha + \beta X_i + \varepsilon_i,$$

where X_i 's are non-stochastic and the ε_i 's are iid with mean 0 and var σ^2 .

$$\begin{aligned} E(\hat{\alpha}) = \alpha & \quad | \quad Var(\hat{\alpha}) = ? \\ E(\hat{\beta}) = \beta & \quad | \quad Var(\hat{\beta}) = ? \\ Cov(\hat{\alpha}, \hat{\beta}) & = ? \end{aligned}$$

In this more general case we consider a linear model of the form

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{k1} + \varepsilon_1 \\ &\vdots \\ Y_n &= \beta_0 + \beta_1 X_{n1} + \cdots + \beta_k X_{nk} + \varepsilon_n \end{aligned}$$

$\varepsilon_1, \dots, \varepsilon_n$ are iid with mean 0 and var σ^2

We write it as

$$Y = X\beta + \varepsilon,$$

where

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{k1} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{kn} \end{bmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

In this case, we have

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T Y \\ E(\hat{\beta}) &= \beta \\ Var(\hat{\beta}) &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

$$E\|\hat{\beta} - \beta\|^2 = \sigma^2 \text{trace}[(X'X)^{-1}] = \sigma^2 \sum_{i=1}^{k+1} \frac{1}{\lambda_i}$$

where λ_i 's are the eigenvalues of $X^T X$.

$$\|\hat{\beta} - \beta\|^2 = (\hat{\beta}_0 - \beta_0)^2 + \dots + (\hat{\beta}_k - \beta_k)^2 = (\hat{\beta} - E(\hat{\beta}))^2$$

In the case of multicollinearity, i.e. when $X^T X$ is near singular, $E(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)$ will be high, so the least square estimates will not be reliable.

A simpler multicollinearity example

Let us consider a simpler example with the model

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i;$$

where the ε_i 's are iid with mean 0 and var σ^2 .

Also assume that all columns are standardized.

$$\begin{array}{ccc} Y_i & X_{1i} & X_{2i} \\ Y_i & X_{1i} & X_{2i} \\ \vdots & \vdots & \vdots \\ Y_n & X_{1n} & X_{2n} \end{array} \quad \begin{array}{l} \bar{Y} = 0 = \bar{X}_1 = \bar{X}_2 \\ S_Y^2 = 1 = S_1^2 = S_2^2 \end{array}$$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = (X'X)^{-1} X'Y$$

$$V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

$$\hat{\beta} = \frac{1}{1 - \rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} \times \begin{bmatrix} \rho Y_1 \\ \rho Y_2 \end{bmatrix}$$

$$= \frac{1}{1 - \rho^2} \begin{Bmatrix} \rho Y_1 - \rho^2 Y_2 \\ \rho Y_2 - \rho^2 Y_1 \end{Bmatrix}$$

$$V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

$$= \frac{\sigma^2}{n(1 - \rho^2)} \begin{bmatrix} 1 - \rho & -\rho \\ -\rho & 1 \end{bmatrix}$$

$$\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{n(1 - \rho)}$$

Note:

$$E(\hat{\beta} - \beta)'(\hat{\beta} - \beta) = \sum_{i=0}^k Var(\hat{\beta}_i)$$

$$Var(1'\hat{\beta}) = 1'\sigma^2(X'X)^{-1}1$$

Now,

$$X'X = \begin{bmatrix} \sum X_{1i}^2 & \sum X_{1i}X_{2i} \\ \sum X_{1i}X_{2i} & \sum X_{2i}^2 \end{bmatrix}$$

$$X = \begin{bmatrix} X_{11} & X_{21} \\ X_{12} & X_{22} \\ X_{13} & X_{23} \\ \vdots & \vdots \\ X_{1n} & X_{2n} \end{bmatrix}$$

$$= \begin{bmatrix} n & n\rho \\ n\rho & n \end{bmatrix}$$

$$= n \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \quad \text{where } \rho = \text{corr}(X_1, X_2).$$

$$X'Y = \begin{bmatrix} \sum X_{1i}Y_i \\ \sum X_{2i}Y_i \end{bmatrix} = n \begin{bmatrix} \rho Y_1 \\ \rho Y_2 \end{bmatrix}, \quad \text{where } \rho_{Y_i} = \text{corr}(Y, X_i), \quad i = 1, 2.$$

If $\rho \approx \pm 1$:

- $Var(\hat{\beta}_1)$ & $Var(\hat{\beta}_2)$ will be large, i.e. $E\|\hat{\beta} - \beta\|^2$ will be large
- Magnitudes of $\hat{\beta}_1$ and $\hat{\beta}_2$ will be high
- Signs of $\hat{\beta}_1$ & $\hat{\beta}_2$ can be misleading.

$$\rho_{y_1} > 0$$

$$\rho_{y_2} > 0$$

SL-I W02 L02

How to take care of multicollinearity?

- **Subset selection:** We can go for forward selection or backward deletion method to get a nested sequence of models and then select one of them using an appropriate Criterion (e.g., adjusted R^2 , Mallows's C_p , AIC, BIC)

Ridge regression

Minimize $\|Y - X\beta\|^2$ subject to $\|\beta\|^2 \leq C$ for some given $C > 0$.

This is equivalent to minimize

$$\|Y - X\beta\|^2 + \lambda(\|\beta\|^2 - C) \quad \text{where } \lambda > 0$$

or

$$\|Y - X\beta\|^2 + \lambda\|\beta\|^2 \quad \text{w.r.t. } \beta.$$

Let

$$\Psi(\beta) = (Y - X\beta)'(Y - X\beta) + \lambda\beta'\beta = Y'Y - \beta'X'Y - Y'X\beta + \beta'X'X\beta + \lambda\beta'\beta.$$

Then

$$\nabla\Psi(\beta) = -2X'Y + 2X'X\beta + 2\lambda\beta,$$

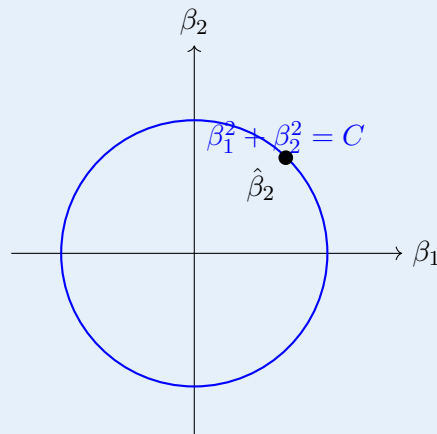
$$\nabla\Psi(\beta) = 0 \implies X'X\beta + \lambda\beta = X'Y \implies (X'X + \lambda I)\beta = X'Y$$

$$\implies \beta = (X'X + \lambda I)^{-1}X'Y.$$

Notation:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{p1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \cdots & x_{pn} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}.$$

Geometric interpretation

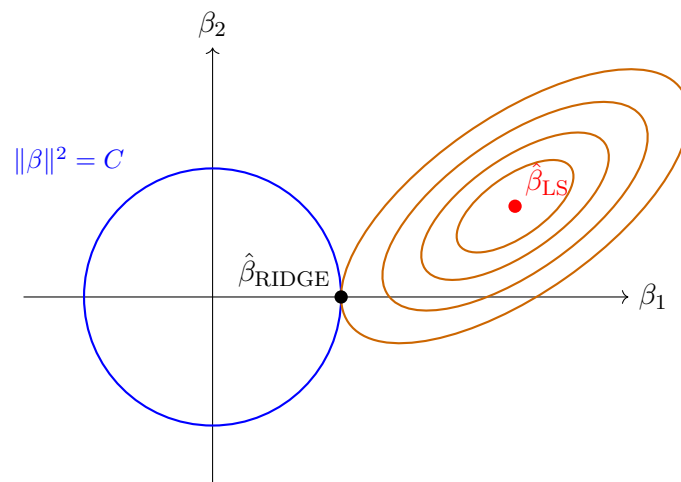


Result:

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{\text{LS}}\|^2 + (\beta - \hat{\beta}_{\text{LS}})'X'X(\beta - \hat{\beta}_{\text{LS}}).$$

The level curves of the residual sum of squares are ellipses centred at $\hat{\beta}_{\text{LS}}$. The ridge estimator is the point where the ellipse first touches the ball $\|\beta\|^2 = C$.

$$\|\hat{\beta}_{\text{RIDGE}}\|^2 = C, \quad \hat{\beta}_{\text{RIDGE}} = (X'X + \lambda I)^{-1}X'Y.$$



For the two-variable illustration:

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i, \quad i = 1, 2, \dots, n.$$

How to choose C (or λ) in practice?

Cross-validation method can be used for this purpose.

$$\min \|Y - X\beta\|^2 \quad \text{s.t.} \quad \beta' \beta \leq C.$$

$$\mathbb{E}(\hat{\beta}_{\text{ridge}}) = (X'X + \lambda I)^{-1}X'X\beta \neq \beta.$$

(The ridge estimator is biased.)

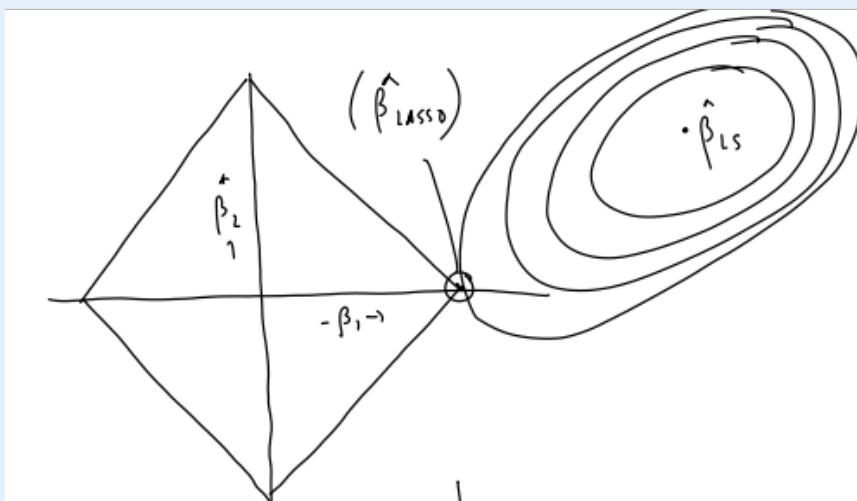
LASSO Regression

(Least Absolute Shrinkage and Selection Operator)

Here we minimize $\|Y - X\beta\|^2$ subject to $\|\beta\|_1 \leq C$ for some $C > 0$.

(For a vector $z \in \mathbb{R}^p$, $\|z\|_1 = |z_1| + |z_2| + \dots + |z_p|$.)

Geometry of the ℓ_1 constraint



$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}_{LS}\|^2 + (\beta - \hat{\beta}_{LS})' X' X (\beta - \hat{\beta}_{LS}).$$

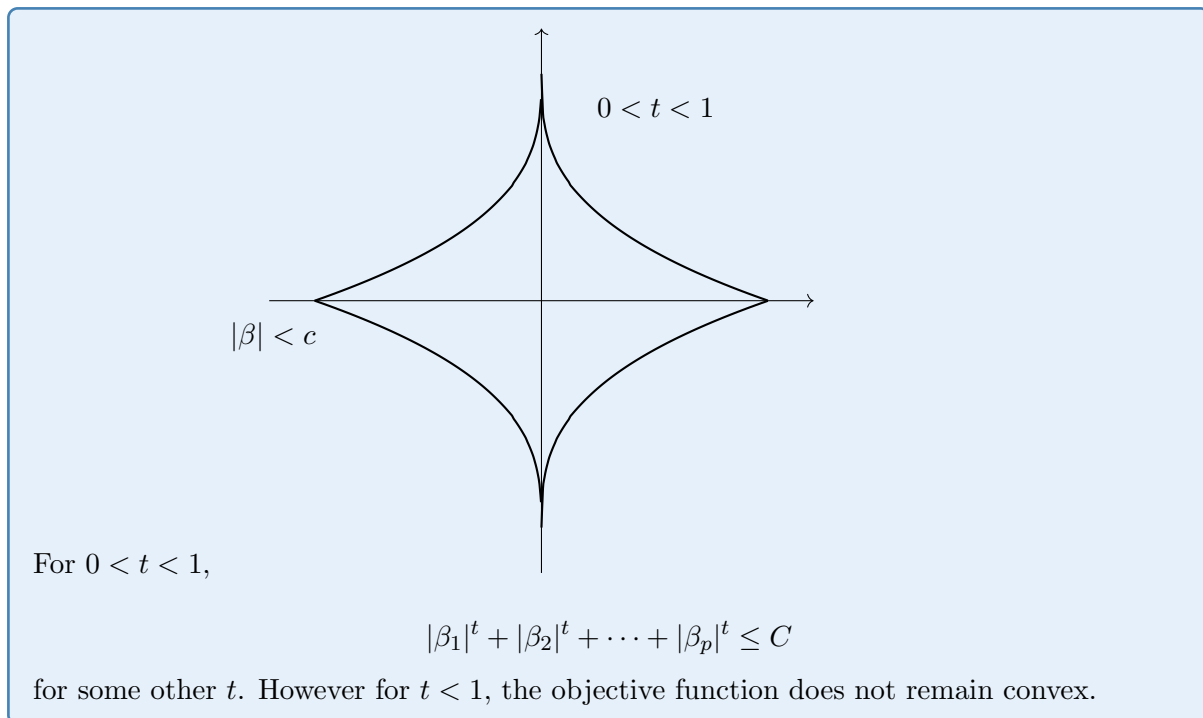
Minimization of

$$\|Y - X\beta\|^2 \quad \text{s.t.} \quad |\beta_1| + \dots + |\beta_p| \leq C$$

is equivalent to minimization of

$$\|Y - X\beta\|^2 + \lambda(|\beta_1| + \dots + |\beta_p| - C).$$

This is a convex function.



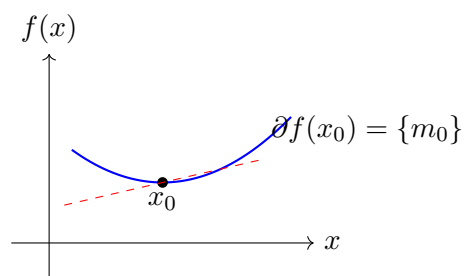
Sub-differential

The sub-differential of a function f at any point x_0 is defined as

$$\partial f(x_0) = \{c : f(x) - f(x_0) \geq c'(x - x_0) \forall x \in \mathbb{R}\}.$$

If f is convex and $f'(x_0)$ exists, then

$$\partial f(x_0) = \{m_0\} \quad \text{where} \quad f'(x_0) = m_0.$$



For $f(x) = |x|$,

$$\partial f(x_0) = \begin{cases} \{1\} & \text{if } x_0 > 0, \\ \{-1\} & \text{if } x_0 < 0, \\ [-1, 1] & \text{if } x_0 = 0. \end{cases}$$

If $0 \in \partial f(x_0)$, then x_0 is a minimizer of $f(\cdot)$.

$$\partial f(x_0) = \{c : f(x) - f(x_0) \geq c'(x - x_0)\}, \forall x \in \mathbb{R}^d.$$

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

Example

$f(x) = \|x\|$, the Euclidean norm

$$= \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}.$$

Show that

$$\partial f(x_0) = \begin{cases} \frac{x_0}{\|x_0\|} & \text{if } x_0 \neq 0, \\ k? & \text{if } x_0 = 0. \end{cases} \quad k = \{c : \|c\| \leq 1\}$$

(The set $\{c : \|c\| \leq 1\}$ is the unit ball in the dual norm.)

LASSO solution (for a special case where $X'X = I$)

Here we minimize

$$\Delta = \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ji} \right)^2 + \lambda (|\beta_1| + |\beta_2| + \cdots + |\beta_p|).$$

If $\beta_k > 0$,

$$\frac{\partial \Delta}{\partial \beta_k} = - \sum_{i=1}^n 2 \left(y_i - \sum_{j=1}^p \beta_j x_{ji} \right) x_{ki} + \lambda.$$

Setting the derivative to zero:

$$\begin{aligned} \frac{\partial \Delta}{\partial \beta_k} = 0 &\implies -2 \sum_i y_i x_{ki} + 2\beta_k n + \lambda = 0 \\ &\implies -2\hat{\beta}_{k,LS} + 2\beta_k + \lambda = 0 \end{aligned}$$

(using $X'X = I$, so $\sum_i x_{ji} x_{ki} = \delta_{jk}$ and the least-squares estimator simplifies accordingly).

More generally the sub-gradient condition is used when some coordinates are zero.

Sub-gradient definition (vector case)

$$\partial f(x_0) = \left\{ c : f(x) - f(x_0) \geq c'(x - x_0) \quad \forall x \in \mathbb{R}^d \right\},$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

$$\|Y - X\beta\|^2 = \sum_i (y_i - \beta_1 x_{1i} - \cdots - \beta_p x_{pi})^2.$$

Since $X'X = I$,

$$\sum_i x_{ji} x_{ki} = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

1 LASSO when $X^\top X = I$

Consider the case $X^\top X = I$.

In this case of LASSO, we minimize

$$\Delta = \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ji} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|. \quad (1)$$

Case $\beta_k > 0$

If $\beta_k > 0$,

$$\frac{\partial \Delta}{\partial \beta_k} = \sum_{i=1}^n \left\{ 2 \left(y_i - \sum_{j=1}^p \beta_j x_{ji} \right) (-x_{ki}) \right\} + \lambda.$$

Setting the derivative to zero:

$$\frac{\partial \Delta}{\partial \beta_k} = 0 \quad \Rightarrow \quad - \sum_{i=1}^n y_i x_{ki} + \sum_{i=1}^n \sum_{j=1}^p \beta_j x_{ji} x_{ki} + \frac{\lambda}{2} = 0.$$

Expanding the second summation,

$$-\hat{\beta}_{k,LS} + \beta_k \sum_{i=1}^n x_{ki}^2 + \beta_2 \sum_{i=1}^n x_{2i} x_{ki} + \cdots + \beta_p \sum_{i=1}^n x_{pi} x_{ki} + \frac{\lambda}{2} = 0.$$

Since $X^\top X = I$, all cross-product terms vanish and $\sum_{i=1}^n x_{ki}^2 = 1$. Hence,

$$\hat{\beta}_{k,LS} = \beta_k + \frac{\lambda}{2} \quad \Rightarrow \quad \beta_k = \hat{\beta}_{k,LS} - \frac{\lambda}{2}.$$

$$\hat{\beta}_{k,LASSO} = \hat{\beta}_{k,LS} - \frac{\lambda}{2} \quad \text{if} \quad \hat{\beta}_{k,LS} > \frac{\lambda}{2} \quad (\beta_k > 0).$$

Case $\beta_k < 0$

Similarly one can show that

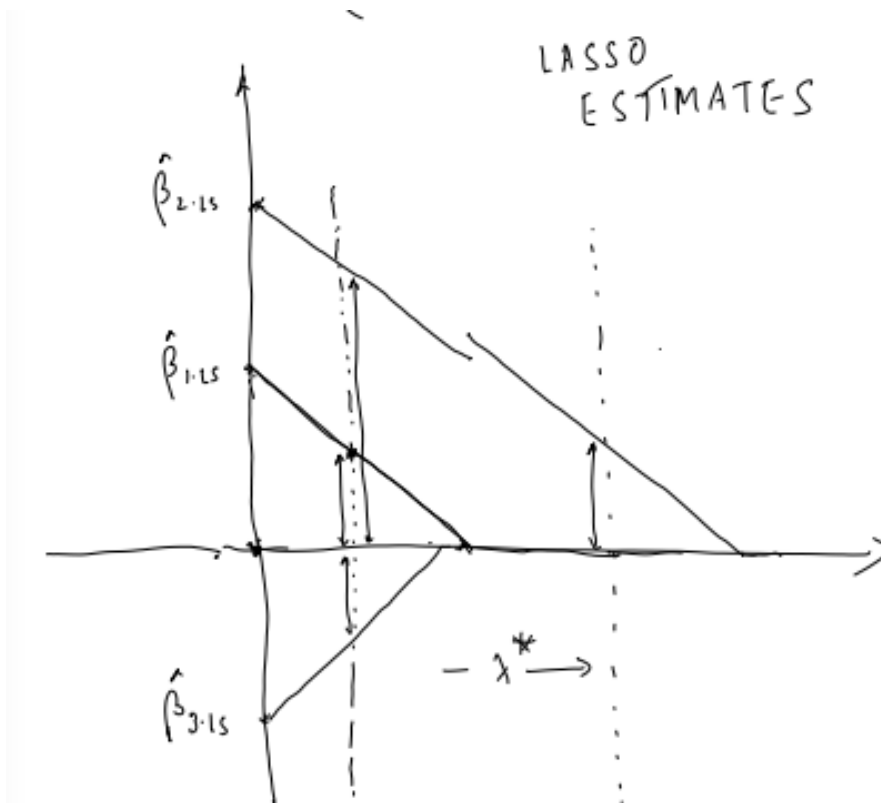
$$\hat{\beta}_{k,LASSO} = \hat{\beta}_{k,LS} + \frac{\lambda}{2} \quad \text{if} \quad \hat{\beta}_{k,LS} < -\frac{\lambda}{2} \quad (\beta_k < 0).$$

Case $\beta_k = 0$

What if $\beta_k = 0$? The condition $0 \in \partial \Delta(\beta_k)$ becomes

$$\hat{\beta}_{k,LS} + c \frac{\lambda}{2} = 0 \quad \Rightarrow \quad |\hat{\beta}_{k,LS}| \leq \frac{\lambda}{2}.$$

$$\begin{aligned} \partial_0(\beta_k) &= \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ji} \right) (-x_{ki}) + c\lambda \quad \text{where } c \in [-1, 1] \\ &= -2\hat{\beta}_{k,LS} + 2\beta_k + c\lambda \in \partial_0(\beta_k) \\ &\Rightarrow \hat{\beta}_{k,LS} + \left[-\frac{\lambda}{2}, \frac{\lambda}{2} \right] \ni 0 \Rightarrow |\hat{\beta}_{k,LS}| \leq \frac{\lambda}{2} \end{aligned}$$



Soft-thresholding rule

Collecting the three cases we obtain the soft-thresholding estimator

$$\hat{\beta}_{k,\text{LASSO}} = \begin{cases} \hat{\beta}_{k,\text{LS}} - \lambda^* & \text{if } \hat{\beta}_{k,\text{LS}} > \lambda^*, \\ \hat{\beta}_{k,\text{LS}} + \lambda^* & \text{if } \hat{\beta}_{k,\text{LS}} < -\lambda^*, \\ 0 & \text{if } |\hat{\beta}_{k,\text{LS}}| < \lambda^*, \end{cases} \quad \lambda^* = \frac{\lambda}{2}.$$

LASSO estimates

The objective function

$$\sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

is the objective function of LASSO.

The value of λ can be chosen using cross-validation (CV).

So when $X^\top X = I$, the use of the penalty parameter λ gradually shrinks the magnitudes of the regression coefficients and eventually makes them zero (deletes the variable from the model) for higher values of λ .

How to compute the LASSO estimates if $X^\top X \neq I$?

We use the **coordinate descent** method for minimisation of the objective function. This is an iterative algorithm. We start with initial estimates $\beta_1^{(0)}, \beta_2^{(0)}, \dots, \beta_p^{(0)}$ and update them iteratively.

$$X = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{p1} \\ x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \cdots & x_{pn} \end{bmatrix}$$

$$\Delta = \sum_{i=1}^n \left(y_i - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + 2\lambda \sum_{j=1}^p |\beta_j|.$$

In order to update the LASSO estimate for β_1 we minimise

$$\sum_{i=1}^n \left(y_i - \beta_1 x_{1i} - \sum_{j=2}^p \beta_j^{(0)} x_{ji} \right)^2 + 2\lambda \left(|\beta_1| + \sum_{j=2}^p |\beta_j^{(0)}| \right).$$

Define the partial residual

$$\sum_{i=1}^n 2(z_i - \beta_1 x_{1i})(-x_{i1}) = -2 \sum_{i=1}^n z_i x_{i1} + 2\beta_1 \sum_{i=1}^n x_{1i}^2$$

Then the problem reduces to

$$\sum_{i=1}^n (z_i - \beta_1 x_{1i})^2 + 2\lambda |\beta_1| \quad (+\lambda \text{const}), \text{ where } z_i = y_i - \sum_{j \neq 1} \beta_j^{(0)} x_{ji}$$

The derivative with respect to β_1 (when $\beta_1 > 0$) is

$$-2 \sum_i z_i x_{i1} + 2\beta_1 \sum_i x_{1i}^2 + 2\lambda.$$

Equating it with zero one gets

$$\beta_1^{(1)} = \frac{\sum z_i x_{i1} - \lambda}{\sum x_{1i}^2} \quad \text{if } \sum z_i x_{i1} > \lambda.$$

$$\beta_1 = \frac{\sum z_i x_{ii} - \lambda}{\sum x_{ii}^2} \quad \text{if } \sum z_i x_{ii} > \lambda.$$

Similarly we can derive the expression for $\beta_1^{(1)}$ in the other two cases. Combining all three cases we obtain

$$\beta_1^{(1)} = \begin{cases} \frac{\sum z_i x_{i1} - \lambda}{\sum x_{1i}^2} & \text{if } \sum z_i x_{i1} > \lambda, \\ \frac{\sum z_i x_{1i} + \lambda}{\sum x_{1i}^2} & \text{if } \sum z_i x_{1i} < -\lambda, \\ 0 & \text{if } |\sum z_i x_{i1}| < \lambda. \end{cases}$$

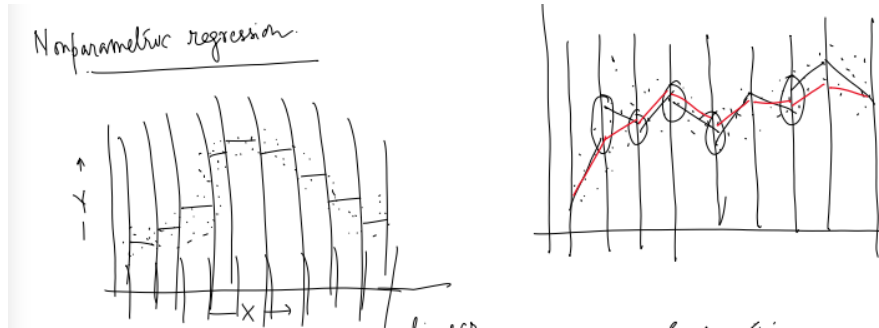
When $X^\top X = I$ we have $\sum x_{1i}^2 = 1$ and

$$\sum z_i x_{1i} = \sum y_i x_{1i} - \sum_{j \neq 1} \beta_j^{(0)} \underbrace{\sum_i x_{ji} x_{1i}}_{=0} = \hat{\beta}_{1, \text{LS}},$$

recovering the soft-thresholding rule of the previous section.

This coordinate-descent method works as long as the non-differentiable part of the objective function is separable.

2 Nonparametric Regression



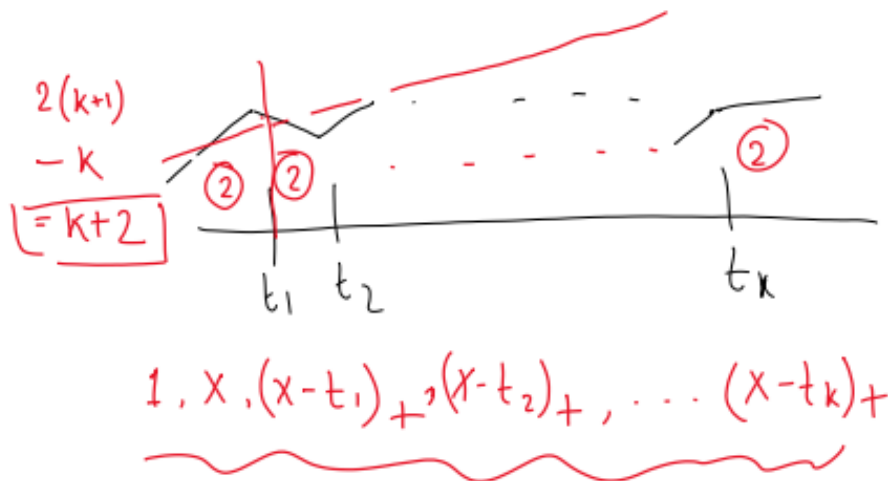
Here we want a piecewise linear estimate which is continuous. We divide the whole range of x into several parts

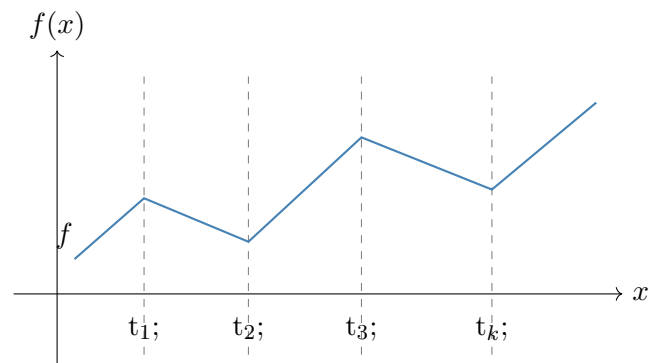
$$(-\infty, t_1), (t_1, t_2), \dots, (t_{k-1}, t_k), (t_k, +\infty)$$

and look for a function which is

- i) linear on each (t_{j-1}, t_j) ,
- ii) continuous at the knots $t_1, \dots, t_k$.

Such a function is called a **linear spline**.





A basis for linear splines is given by the truncated-power functions

$$1, \quad x, \quad (x - t_1)_+, \quad (x - t_2)_+, \quad \dots, \quad (x - t_k)_+,$$

where the positive-part function is defined by

$$(x - t)_+ = \begin{cases} x - t & \text{if } x > t, \\ 0 & \text{if o.w.,} \end{cases} \quad \text{i.e.} \quad (x - t)_+ = \max\{0, x - t\}.$$

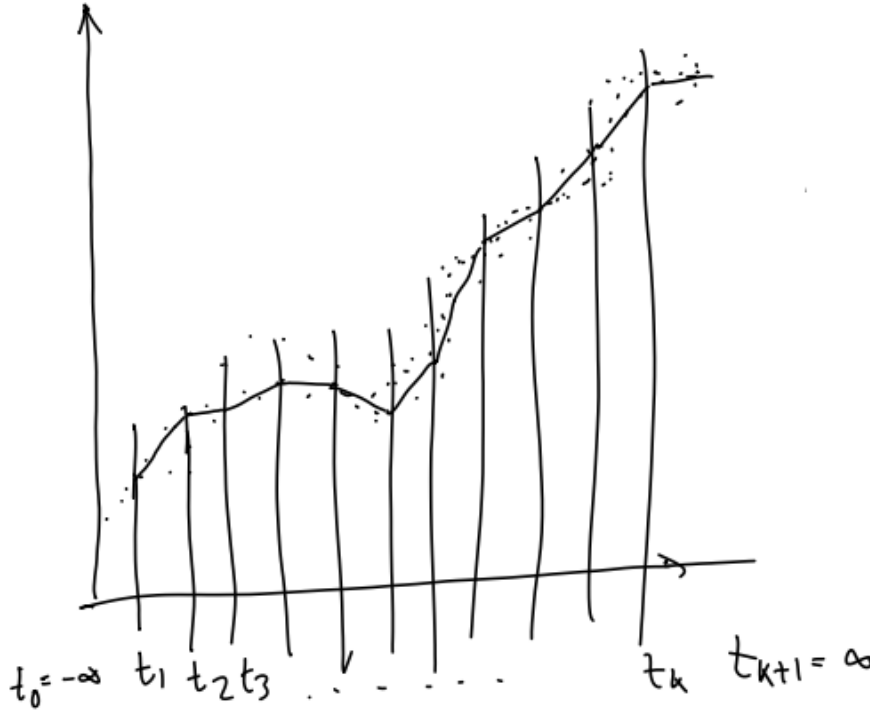
A general linear spline may therefore be written

$$f(x) = \beta_0 + \beta_1 x + \sum_{j=1}^k \beta_{j+1} (x - t_j)_+.$$

End of notes

1 Linear Splines and Truncated Power Basis

The collection of functions



$$\{1, x, (x - t_1)_+, (x - t_2)_+, \dots, (x - t_k)_+\}$$

forms a basis for the class of functions we are interested in.

Now any function f which is linear in the interval (t_{i-1}, t_i) for all i and continuous at the t_i 's can be written as

$$f(x) = \alpha_0 + \alpha_1 x + \alpha_2 (x - t_1)_+ + \alpha_3 (x - t_2)_+ + \dots + \alpha_{k+1} (x - t_k)_+.$$

In L.S. regression, we minimize

$$\sum (Y_i - f(x_i))^2.$$

So, here we would like to minimize

$$\sum [Y_i - \alpha_0 - \alpha_1 x_i - \alpha_2 (x_i - t_1)_+ - \dots - \alpha_{k+1} (x_i - t_k)_+]^2.$$

In matrix form

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_1 & (x_1 - t_1)_+ & \dots & (x_1 - t_k)_+ \\ 1 & x_2 & (x_2 - t_1)_+ & \dots & (x_2 - t_k)_+ \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & (x_n - t_1)_+ & \dots & (x_n - t_k)_+ \end{bmatrix}, \quad \beta = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{k+1} \end{bmatrix}.$$

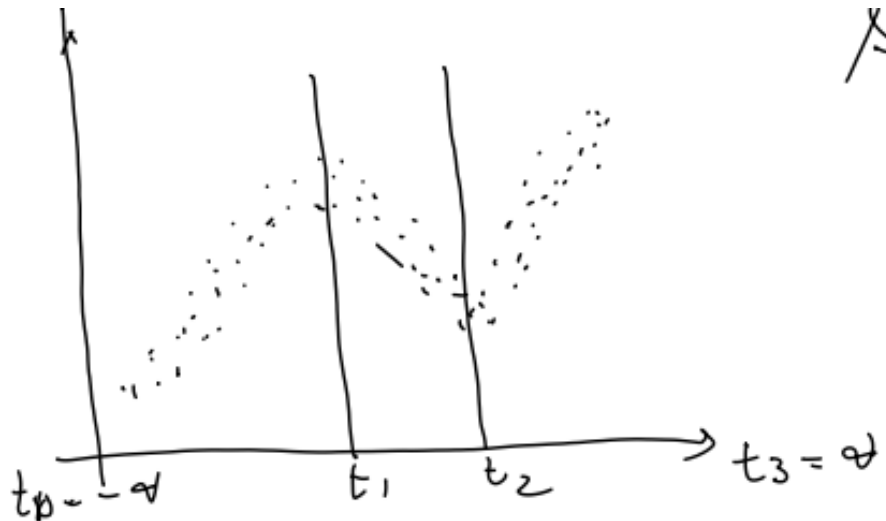
The ordinary least-squares estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

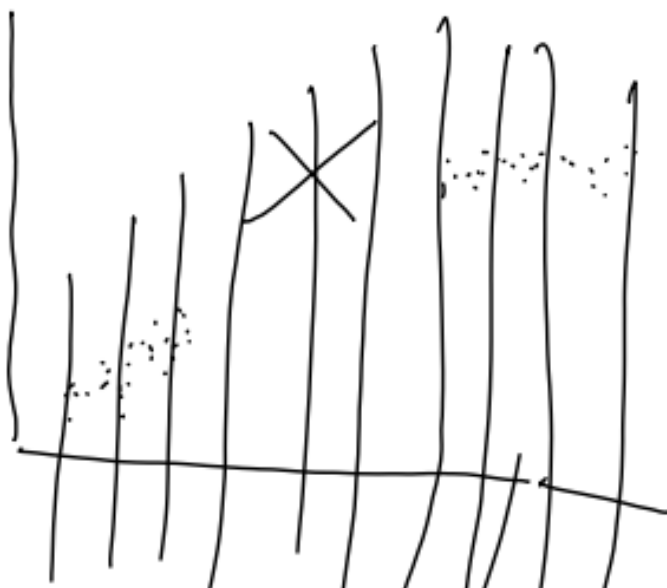
Questions about knots

- How many knots (what is the value of k)?
- Where should we place these knots?

Sometimes the scatter plot of the data set gives you an idea.

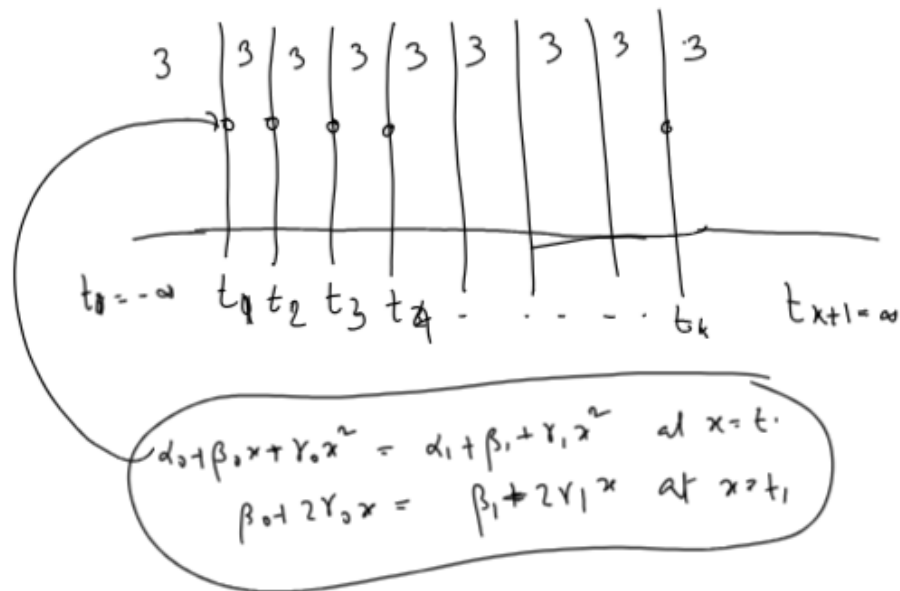


- We can start with sufficiently large number of knots and then use the backward deletion method to get a nested sequence of models. We use appropriate criterion (e.g. adjusted R^2) to select one of these models.
- Taking equi-spaced knots is not always a good option. It is better to place the knots based on order statistics.



Degrees of freedom counting:

$$3(k+1) - 2k = k + 3.$$



The collection of functions

$$\{1, x, x^2, \dots\}$$

2 Quadratic & Cubic Splines; Multivariate Extension

Similarly, for cubic splines we can consider knots at $t_1, t_2, \dots, t_k$. We can consider the basis

$$\{1, x, x^2, x^3, (x - t_1)_+^3, (x - t_2)_+^3, \dots, (x - t_k)_+^3\}.$$

This forms a basis for the quadratic/cubic splines.

This basis is called **power spline basis** or **truncated power spline basis**.

Indeed one can use other basis functions (e.g. B-spline basis) for regression.

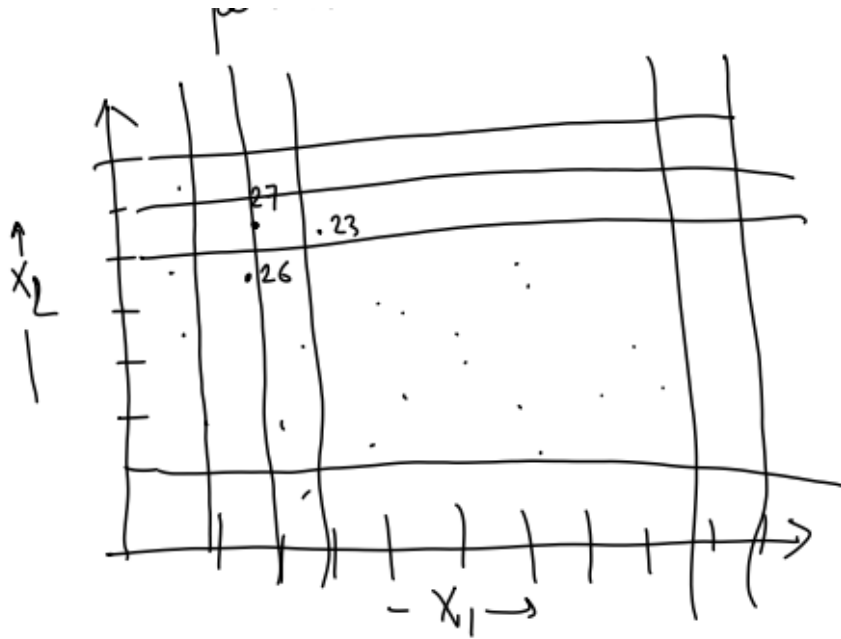
More than one predictor

Now consider the case when there are more than two (or more) predictor variables.

In two dimensions we can consider the basis

$$1, x_1, (x_1 - t_{11})_+, \dots, (x_1 - t_{1k})_+, \\ x_2, (x_2 - t_{21})_+, \dots, (x_2 - t_{2k})_+$$

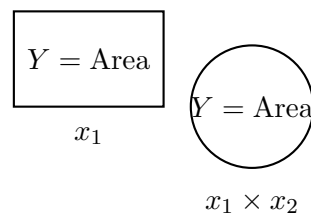
and go for linear regression as before.



If we do that then basically we are assuming that f is of the form

$$\begin{aligned} f(x_1, x_2) &= \alpha_0 + \alpha_1 x_1 + \alpha_2 (x_1 - t_{11})_+ + \cdots + \alpha_{k+1} (x_1 - t_{1k})_+ \\ &\quad + \beta_1 x_2 + \beta_2 (x_2 - t_{21})_+ + \cdots + \beta_{k+1} (x_2 - t_{2k})_+ \\ &= f_1(x_1) + f_2(x_2). \end{aligned}$$

So, here we are assuming an **additive model**.



However, this method won't work if we have functions like $f(x_1, x_2) = x_1 x_2$ (non additive models):

$$\begin{aligned} f(x_1, x_2) &\neq f_1(x_1) + f_2(x_2), \\ x_1 x_2 &\neq f_1(x_1) + f_2(x_2). \end{aligned}$$

But we can rewrite the product as a sum of functions of linear projections:

$$\begin{aligned} x_1 x_2 &= \left(\frac{x_1 + x_2}{2} \right)^2 - \left(\frac{x_1 - x_2}{2} \right)^2 \\ &= f_1(\alpha_1^\top x) + f_2(\alpha_2^\top x), \end{aligned}$$

where

$$\begin{aligned} \alpha_1 &= \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad \alpha_1^\top x = \frac{x_1 + x_2}{\sqrt{2}}, \quad f_1(t) = \frac{t^2}{2}, \\ \alpha_2 &= \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right), \quad \alpha_2^\top x = \frac{x_1 - x_2}{\sqrt{2}}, \quad f_2(t) = -\frac{t^2}{2}. \end{aligned}$$

Additive model:

$$f(x_1, x_2) = f_1(x_1) + f_2(x_2).$$

In general

$$f(x_1, x_2) = \sum_{k=1}^M f_k(\alpha_k^\top x).$$

3 Projection Pursuit Regression & Single-Layer Perceptron

So instead of considering an additive model, one can assume that the underlying regression function f is of the form

$$f(x_1, \dots, x_p) = \sum_{k=1}^M f_k(\alpha_k^\top x), \quad \alpha_k \in \mathbb{R}^p, \quad x = (x_1, \dots, x_p), \quad M \in \mathbb{N}.$$

This type of model is known as the **projection pursuit model**.

- How to estimate the α_k 's, f_k 's and M ?

Single-layer perceptron

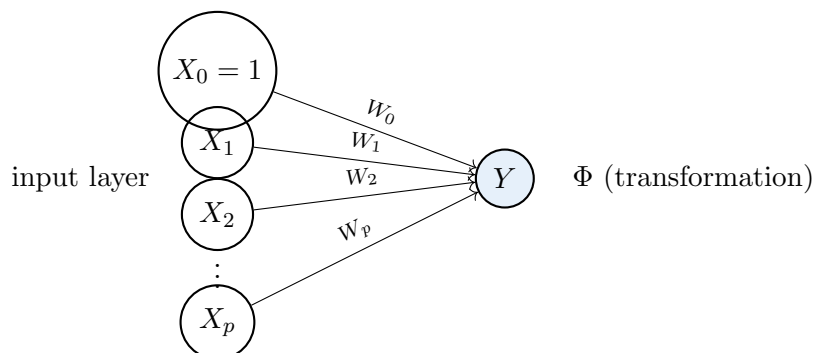
Input–output relationship:

$$Y = \Phi\left(W_0 + \sum_{i=1}^p W_i X_i\right).$$

In a single-layer perceptron model, we have the input–output relationship given by

$$Y = \Phi\left(W_0 + \sum W_i X_i\right).$$

This may be viewed as a generalized linear model.



For $\Phi(t) = t$ we get back the linear model.

Often

$$\Phi(t) = \frac{1}{1 + e^{-t}}$$

is used in the case of classification. We called it the **logistic function** / **sigmoid function**.

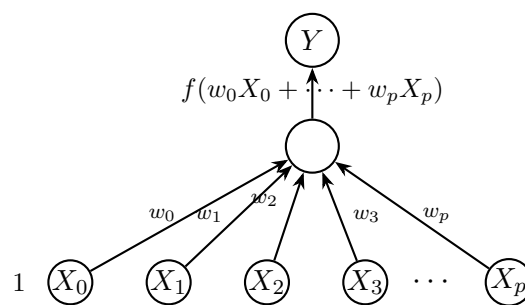
Classification rule:

$$\begin{aligned} Y &\leq C \quad \text{or} \quad Y > C, \\ \Phi(W_0 + \sum W_i X_i) &\leq C \quad \text{or} \quad > C, \\ W_0 + \sum W_i X_i &\leq K_0 \quad \text{or} \quad > K_0. \end{aligned}$$

W04-L01

August 10, 2026

Single Layer Perceptron



Input layer

Single layer perceptron

Input-output relationship

$$Y = f\left(w_0 + \sum_{i=1}^p w_i X_i\right)$$

If $f(t) = t$, then we get the linear model.

Usually, f is known and it is an increasing function / non-decreasing function.

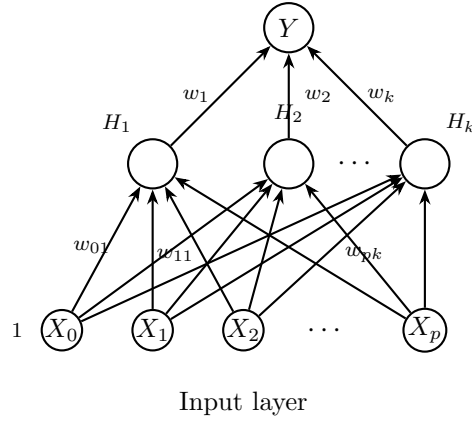
$$\text{example : } \begin{cases} f(t) = \frac{1}{1 + e^{-t}} & \text{Sigmoid} \\ = \max\{0, t\} & \text{ReLU} \end{cases}$$

In general, this gives us a model which can be viewed as a generalized linear model.

Multilayer Perceptron Model

(with one hidden layer)

In order to capture the complex relationship between input and output variables, sometimes we use a complex model, where one or more hidden layers are introduced between the input and the output layers.



Multilayer perceptron model (with one hidden layer)

$$H_j = f_j \left(w_{0j} + \sum_{i=1}^p w_{ij} X_i \right), \quad j = 1, 2, \dots, k$$

$$Y = f \left(w_0 + \sum_{j=1}^k w_j H_j \right)$$

In general,

$$Y = f \left\{ w_0 + \sum_{j=1}^k w_j f_j \left(w_{0j} + \sum_{i=1}^p w_{ij} X_i \right) \right\}$$

If $f(t) = t$,

$$Y = w_0 + \sum_{j=1}^k w_j f_j(\alpha'_j X)$$

$$\alpha_j = (w_{0j}, w_{1j}, \dots, w_{pj}), \quad X = (X_0, X_1, \dots, X_p)$$

i.e. equivalent to write

$$Y = \sum_{j=1}^k f_j^*(\alpha'_j X)$$

which is the projection pursuit model.

Using a multi-layer perceptron model with sigmoid transformations at the hidden layer, one can approximate any continuous function defined on a compact domain to any desired level of accuracy.

$$\sup_x |f(x) - f_{\text{MLP}}(x)| < \varepsilon \quad \text{for any given } \varepsilon > 0$$

provided there are enough neurons in the hidden layer (Hornik, Stinchcombe & White (1989)).

Projection Pursuit Model

Here we are looking for a model of the form

$$f(x_1, \dots, x_p) = \sum_{j=1}^k f_j(\alpha'_j X)$$

and we need to estimate the f_j 's and the α_j 's.

First consider the case $k = 1$.

$$\begin{array}{cccccc} y_1 & x_{11} & x_{21} & \cdots & x_{p1} \\ y_2 & x_{12} & x_{22} & \cdots & x_{p2} \\ \vdots & \vdots & \vdots & & \vdots \\ y_n & x_{1n} & x_{2n} & \cdots & x_{pn} \end{array}$$

How we want to find α_1 and f_1 :

$$\sum_{i=1}^n (y_i - f_1(\alpha'_1 x_i))^2$$

gets minimized.

We can use an iterative method for this purpose.

Note that given any α_1 , we can compute

$$z_1 = \alpha'_1 x_1, \dots, z_n = \alpha'_1 x_n$$

and fit f_1 (approximate f_1) using splines.

How to update α_1 ?

Let $\alpha_1^{(0)}$ be the initial choice of α_1 and $f_1^{(0)}$ be the current estimate of f_1 .

$$\begin{aligned} \sum [y_i - f_1(\alpha'_1 x_i)]^2 &\approx \sum [y_i - f_1^{(0)}(\alpha'_1 x_i)]^2 \\ &\approx \sum \left[y_i - f_1^{(0)}(\alpha_1^{(0)'} x_i) - (\alpha_1 - \alpha_1^{(0)})' \nabla f_1^{(0)}(\alpha_1^{(0)'} x_i) \Big|_{\alpha_1 = \alpha_1^{(0)}} \right]^2 \\ &= \sum [y_i^* - (\alpha_1 - \alpha_1^{(0)})' \delta_i^*]^2 \end{aligned}$$

$$\begin{aligned} \Psi(\alpha) &\approx \Psi(\alpha_0) + (\alpha - \alpha_0)' \Psi'(\alpha) \Big|_{\alpha = \alpha_0} \\ \Psi(\alpha) &= \Psi(\alpha_0) + (\alpha - \alpha_0)' \nabla \Psi(\alpha) \Big|_{\alpha = \alpha_0} \end{aligned}$$

We can find the minimizer and that can be used as $\alpha_1^{(1)}$.

We can proceed this way to get the final estimates of α_1 and f_1 .

If $K > 1$ (e.g. assume that $K = 2$):

After getting α_1 and f_1 , we redefine the response variable

$$\tilde{y} = y - f_1(\alpha'_1 X)$$

and proceed similarly.

$$\sum (y_i - f_1(\alpha'_1 x_i) - f_2(\alpha'_2 x_i))^2$$

$$\sum (\tilde{y}_i - f_2(\alpha'_2 x_i))^2$$

We stop when the reduction in RSS(Residual sum of squares) is less than a user-defined threshold.

Other Estimates for Regression Function

Nearest Neighbor Method

X	5	7	8	8.5	9.7	10.6	11.6	13
Y	11	13	14	15	14	13	12	10

(Example with $k = 3$, $\hat{y} = 14$)

$$\hat{f}_k(x) = \frac{\sum_{x_i \in N_k(x)} y_i}{k}$$

where $N_k(x)$ is the collection of k -neighbors of x .

This nearest neighbor estimate is piecewise constant.

This can be easily generalized to higher dimension.

In order to get the consistency of k -VN estimator, we need

$$k \rightarrow \infty \quad \text{and} \quad \frac{k}{n} \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty$$

(Stone, 1977).

Weighted Nearest Neighbor Method

Instead of nearest neighbor, one can use the weighted nearest neighbor method. In that case, the estimator is defined as

$$\hat{f}_{k\text{-WNN}}(x) = \frac{\sum_{i: x_i \in N_k(x)} w(\|x - x_i\|) Y_i}{\sum_{i: x_i \in N_k(x)} w(\|x - x_i\|)},$$

where $w : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a non-increasing function.

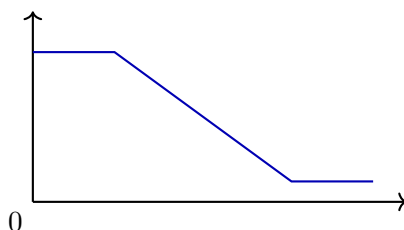


Illustration of a non-increasing weight function w

Lecture 8

August 12, 2026

W04-L02

k -Nearest Neighbour Estimator

$$\hat{f}_k(x) = \frac{\sum_{i: x_i \in N_k(x)} Y_i}{k}$$

For the large sample consistency of this estimator we need $k \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. This is a piecewise constant function; not continuous.

Instead of k -NN, one can think about using a weighted k -NN estimator. This is given by

$$\hat{f}_{k\text{-WNN}}(x) = \frac{\sum_{i: x_i \in N_k(x)} \omega(\|x - x_i\|) Y_i}{\sum_{i: x_i \in N_k(x)} \omega(\|x - x_i\|)}$$

where $\omega : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a non-increasing / decreasing function.

Even if $\omega(\cdot)$ is a continuous function, $\hat{f}_{k\text{-WNN}}$ may not be a continuous function unless $k = n$. For $k = n$, we have the following expression for the estimator

$$\hat{f}(x) = \frac{\sum_{i=1}^n \omega(\|x - x_i\|) Y_i}{\sum_{i=1}^n \omega(\|x - x_i\|)}$$

This is continuous if $\omega(\cdot)$ is a continuous function.

Choice of ω : ω should be decreasing / non-increasing function.

One popular choice is the Gaussian weight function

$$\omega(t) = e^{-\frac{1}{2}t^2}$$

For this choice of the weight function, we have

$$\hat{f}(x) = \frac{\sum_{i=1}^n e^{-\frac{1}{2}\|x-x_i\|^2} Y_i}{\sum_{i=1}^n e^{-\frac{1}{2}\|x-x_i\|^2}}$$

In order to make adjustments for the scale, one can think of using an estimator of the form

$$\hat{f}(x) = \frac{\sum_{i=1}^n e^{-\frac{1}{2h^2}\|x-x_i\|^2} Y_i}{\sum_{i=1}^n e^{-\frac{1}{2h^2}\|x-x_i\|^2}} = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)}$$

where

$$K(t) = e^{-\frac{1}{2}t't} = e^{-\frac{1}{2}\|t\|^2}.$$

Instead of using $K(t) = e^{-\frac{1}{2}t't}$, one can use other kernel functions as well.

This type of estimator is known as the kernel regression estimator or the Nadaraya–Watson estimator.

$K(\cdot)$ is called the kernel function and h is the associated smoothing parameter known as the bandwidth.

Nadaraya–Watson Estimator

NW Est.:

$$\hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)}$$

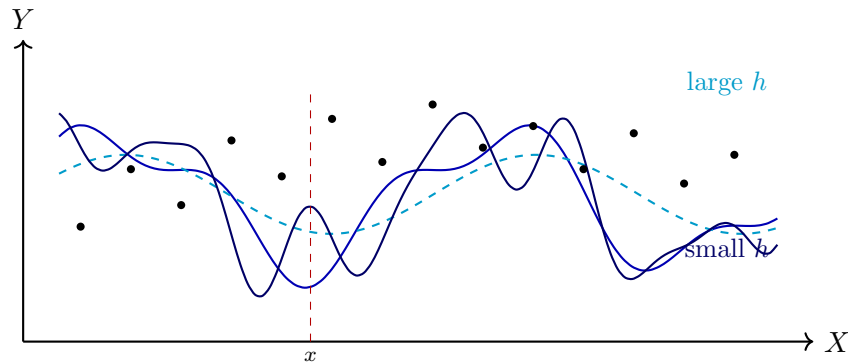


Illustration of the effect of bandwidth h : large h yields a smoother estimate; small h follows the data more closely.

Assume that

$$K(b) = e^{-\frac{1}{2}b^2}.$$

Let $x_1, x_2 \in \mathbb{R}^p$ and $x \in \mathbb{R}^p$.

$$\frac{K\left(\frac{x-x_1}{h}\right)}{K\left(\frac{x-x_2}{h}\right)}.$$

If $\|x - x_1\| < \|x - x_2\|$, then

$$\frac{K\left(\frac{x-x_1}{h}\right)}{K\left(\frac{x-x_2}{h}\right)} = \frac{e^{-\frac{1}{2h^2}\|x-x_1\|^2}}{e^{-\frac{1}{2h^2}\|x-x_2\|^2}} = e^{-\frac{1}{2h^2}\{\|x-x_1\|^2 - \|x-x_2\|^2\}} \uparrow \infty \quad \text{as } h \downarrow 0.$$

Or in other words,

$$\frac{K\left(\frac{x-x_2}{h}\right)}{K\left(\frac{x-x_1}{h}\right)} \downarrow 0 \quad \text{as } h \downarrow 0.$$

Data points: $x_1, \dots, x_n$

Query point: x

Let x_k be the nearest neighbour of x .

$$\lim_{h \downarrow 0} \hat{f}_h(x) = \frac{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)} = Y_k.$$

Cross-validation method can be used to choose the value of h .

Data

$$\{x_1, x_2, \dots, x_n\}, \quad \{Y_1, Y_2, \dots, Y_n\}.$$

$$\min_a \sum (Y_i - a)^2 \longrightarrow a = \bar{Y}.$$

$$\hat{f}(x) = a, \quad \min_a \sum (Y_i - a)^2 K\left(\frac{x-x_i}{h}\right).$$

If we minimise $\sum (Y_i - a)^2$ w.r.t. a , we get $\hat{a} = \bar{Y}$.

If we minimise $\sum (Y_i - a)^2 K\left(\frac{x-x_i}{h}\right)$ w.r.t. a , we get

$$\hat{a} = \frac{\sum K\left(\frac{x-x_i}{h}\right) Y_i}{\sum K\left(\frac{x-x_i}{h}\right)} \longrightarrow \text{Kernel regression estimator (Nadaraya-Watson estimator)}.$$

Locally Constant and Local Linear Estimators

Because of this the Nadaraya-Watson estimator is often called the **locally constant estimator**.

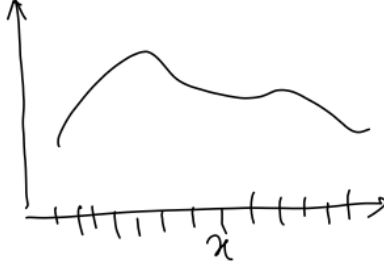
How to get a local linear estimator?

For finding the local linear estimate at x , we shall minimise

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 K\left(\frac{x - X_i}{h}\right) \quad (\text{in the univariate case}).$$

Let $(\hat{\beta}_0, \hat{\beta}_1)$ be the minimiser. Then the local linear estimate of the regression function at x is given by

$$\hat{f}_{LL,h}(x) = \hat{\beta}_0 + \hat{\beta}_1 x.$$



Local linear estimation (alternative method)

We assume that the regression function is linear near x :

$$Y = \beta_0 + \beta_1 X = Y_0 + Y_1(X - x).$$

Minimise

$$\sum_{i=1}^n (Y_i - Y_0 - Y_1(X_i - x))^2 K\left(\frac{x - X_i}{h}\right).$$

Let $(\hat{Y}_0, \hat{Y}_1)$ be the minimiser. Then

$$\hat{Y} \text{ at } x = \hat{Y}_0.$$

Local Polynomial Regression

For finding $\hat{f}(x)$, we minimise

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i - \beta_2 X_i^2 - \cdots - \beta_p X_i^p)^2 K\left(\frac{x - X_i}{h}\right)$$

to get $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$.

The local polynomial estimate of f at x is given by

$$\hat{f}_{LP}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \cdots + \hat{\beta}_p x^p.$$

Alternative method

Minimise

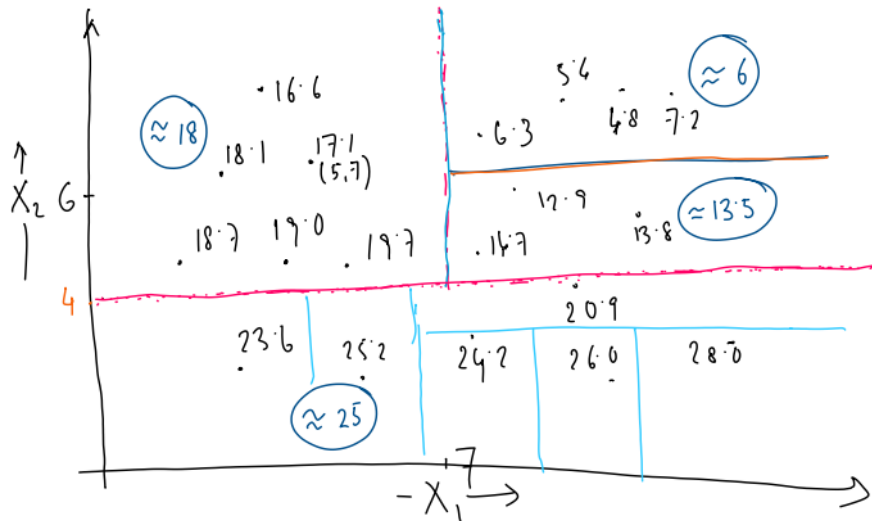
$$\sum_{i=1}^n \left\{ Y_i - Y_0 - Y_1(X_i - x) - Y_2(X_i - x)^2 - \cdots - Y_p(X_i - x)^p \right\}^2 K\left(\frac{x - X_i}{h}\right).$$

If $(\hat{Y}_0, \hat{Y}_1, \dots, \hat{Y}_p)$ is the minimiser, the local polynomial estimate of f at x is given by

$$\hat{f}_{\text{LP},h}(x) = \hat{Y}_0.$$

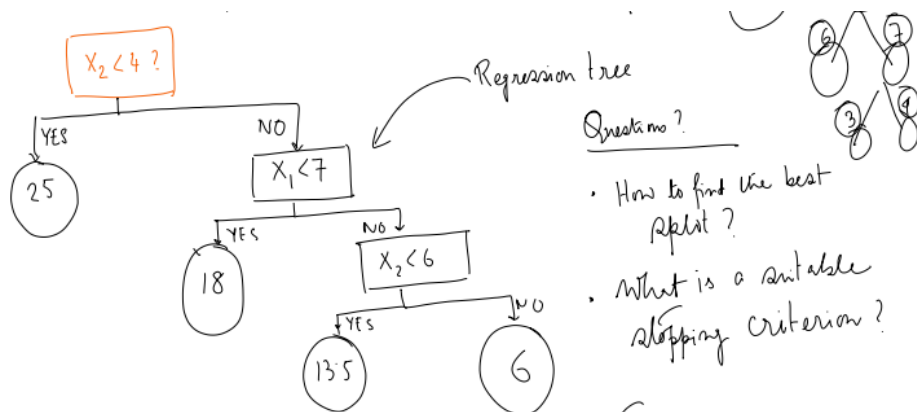
Lecture 9

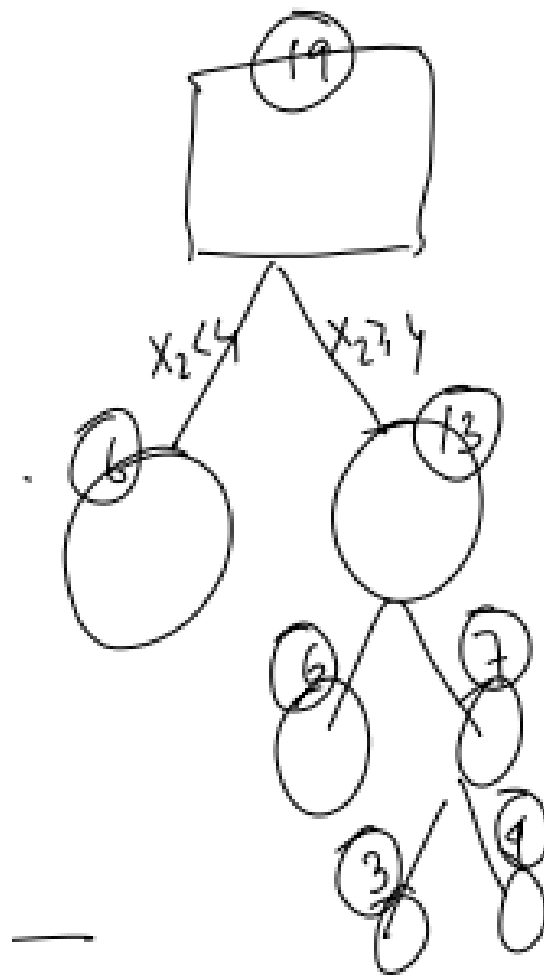
Piecewise constant estimate of a regression function where the pieces will be formed adaptively



Dataset example

X_1	X_2	Y
5	7	17.1
$\vdots$	$\vdots$	$\vdots$



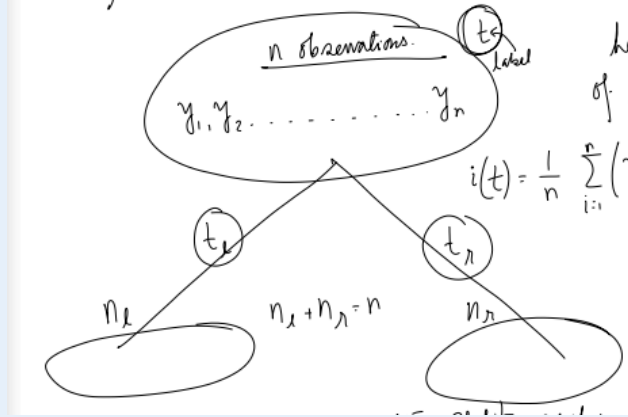


Questions:

- How to find the best split?
- What is a suitable stopping criterion?

Here at each stage, one should try to find the split such that the observations in each of the newly formed sub-regions are more homogeneous compared to the observations in the whole region.

The heterogeneity of a node can be calculated as the variance of the observations (var of the Y -values) belonging to that node.



Heterogeneity (impurity) of the node t

$$i(t) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_t)^2, \quad \text{where } \bar{y}_t = \frac{1}{n} \sum_{i=1}^n y_i$$

For the left and right child nodes:

$$i(t_L) = \frac{1}{n_L} \sum_{i: \text{obs. belongs to } t_L} (y_i - \bar{y}_{t_L})^2$$

$$i(t_R) = \frac{1}{n_R} \sum_{i: \text{obs. belongs to } t_R} (y_i - \bar{y}_{t_R})^2$$

Ideally one should choose the split so that $i(t_L)$ and $i(t_R)$ are much smaller than $i(t)$.

We define the **reduction in impurity**

$$\Delta i(t) = i(t) - \left\{ \frac{n_L}{n} i(t_L) + \frac{n_R}{n} i(t_R) \right\}.$$

Naturally, one would prefer the split for which the reduction is maximum.

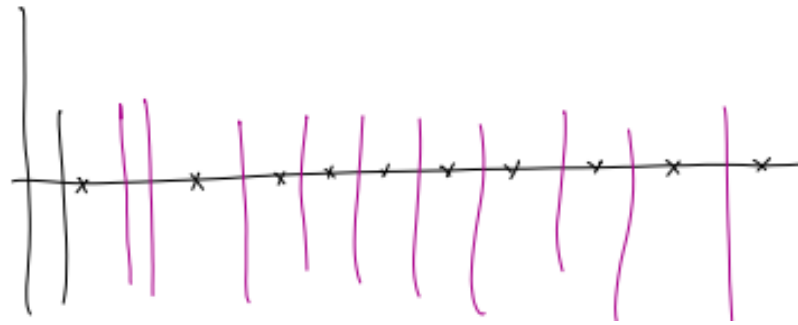
Result: For any split of the node t , $\Delta i(t) \geq 0$.

$$\begin{aligned} nS^2 &= n_L S_L^2 + n_R S_R^2 + n_L d_L^2 + n_R d_R^2 \\ &= n_L S_L^2 + n_R S_R^2 + n_L (\bar{y}_L - \bar{y})^2 + n_R (\bar{y}_R - \bar{y})^2 \end{aligned}$$

$$ni(t) = n_L i(t_L) + n_R i(t_R) + n_L (\bar{y}_L - \bar{y})^2 + n_R (\bar{y}_R - \bar{y})^2$$

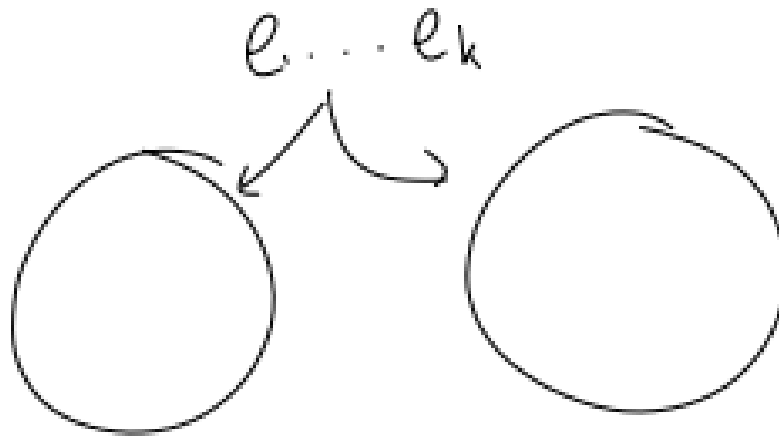
$$i(t) - \left\{ \frac{n_L}{n} i(t_L) + \frac{n_R}{n} i(t_R) \right\} = \frac{n_L}{n} (\bar{y}_L - \bar{y})^2 + \frac{n_R}{n} (\bar{y}_R - \bar{y})^2$$

$$\approx \frac{n_L n_R}{n^2} (\bar{y}_L - \bar{y}_R)^2 \geq 0.$$



If X_1 is a continuous r.v., then there are $n - 1$ many possible splits. We can calculate $\Delta i(t)$ for each of them and consider the best one.

(Discrete-valued var or ordinal var can be handled in the same way. There the no. of candidate splits = # of distinct values $- 1$.)



If the predictor variable is categorical (suppose it has K categories $e_1, e_2, \dots, e_K$), then we have

$$\frac{2^K - 2}{2}$$

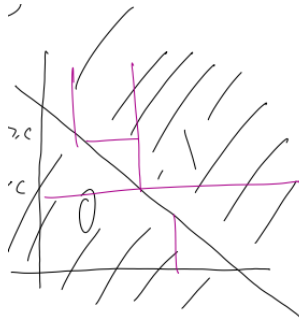
possible candidate splits and one needs to choose the best one.

After finding the best split based on each of the predictors, we finally choose the one which gives the maximum reduction.

Advantages

- It can handle several types of variables.
- Can be conveniently used for high-dimensional data.
- Several splits may be needed to approximate a simple function

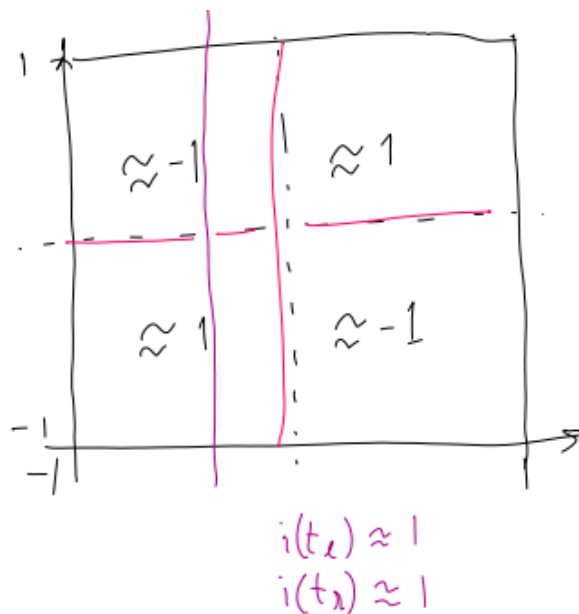
$$f(x) = \begin{cases} 1 & \text{if } aX_1 + bx_2 \geq c, \\ 0 & \text{if } aX_1 + bx_2 < c. \end{cases}$$



- The regression tree estimate usually does have high variance (a slight change in the data may lead to an altogether different type of tree).

When to stop?

- $\Delta i(t) < \varepsilon$. We can define threshold ε .
- The no. of obs. in a region (node) is less than a threshold (say 5, or 1% of the training data).
- If $i(t) < \varepsilon_0$, we may not go for further splitting.



Illustrative example

$$X_1, X_2 \sim U[-1, 1] \times U[-1, 1], \quad Y = f(X_1, X_2) + \varepsilon, \quad \varepsilon \sim N(0, 0.01)$$

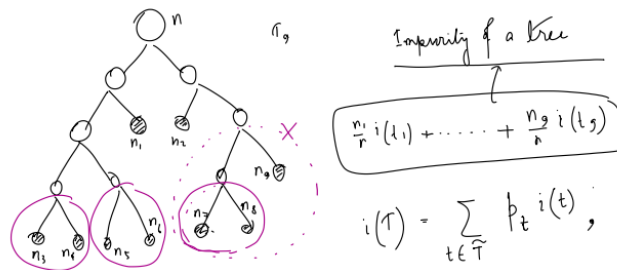
$$f(x_1, x_2) = \begin{cases} +1 & \text{if } x_1 x_2 > 0, \\ -1 & \text{if } x_1 x_2 \leq 0. \end{cases}$$

In this case $i(t) \approx 1$, $i(t_L) \approx 1$, $i(t_R) \approx 1$.

Lecture 10

Pruning

After growing a large-sized tree (using an appropriate stopping criterion), we usually do pruning to get smaller trees.



Impurity of a tree

$$i(T) = \sum_{t \in \tilde{T}} p_t i(t),$$

where $\tilde{T}$ denotes the collection of leaf nodes in T , and

p_t = proportion of observations in the node t .

During pruning, at every stage, we merge two leaf nodes so that the resulting tree will have one leaf node less than the previous one.

We always choose those two nodes whose merging leads to the least increment in the impurity of the tree.

In this way, we get a nested sequence of trees indexed by the no. of leaf nodes

$$T_M \succ T_{M-1} \succ T_{M-2} \succ \cdots \succ T_2 \succ T_1 \text{ (no split).}$$

BFOS (1984)

Breiman, Friedman, Olshen, Stone (1984) proposed to use cost-complexity & cross-validation to choose one of these models and hence the right-sized tree.

They proposed to use a cost function

$$R_\alpha(T) = R(T) + \alpha |\tilde{T}|,$$

where

- $R(T)$ = residual sum of squares,
- $|\tilde{T}|$ = no. of leaf nodes in T ,
- $\alpha > 0$ is the penalty parameter.

Choice of α	Best model
$(0, \alpha_0)$	T_M
(α_0, α_1)	T_{M-1}
(α_1, α_2)	T_{M-2}
$\vdots$	$\vdots$
(α_n, ∞)	T_1 (no split)

$$\begin{array}{c|c}
 \underline{T_M} & \underline{T_k} \quad (k < M) \\
 R(T_M) & R(T_k) \\
 + \alpha M & + \alpha k
 \end{array}$$

T_k is better than T_M if

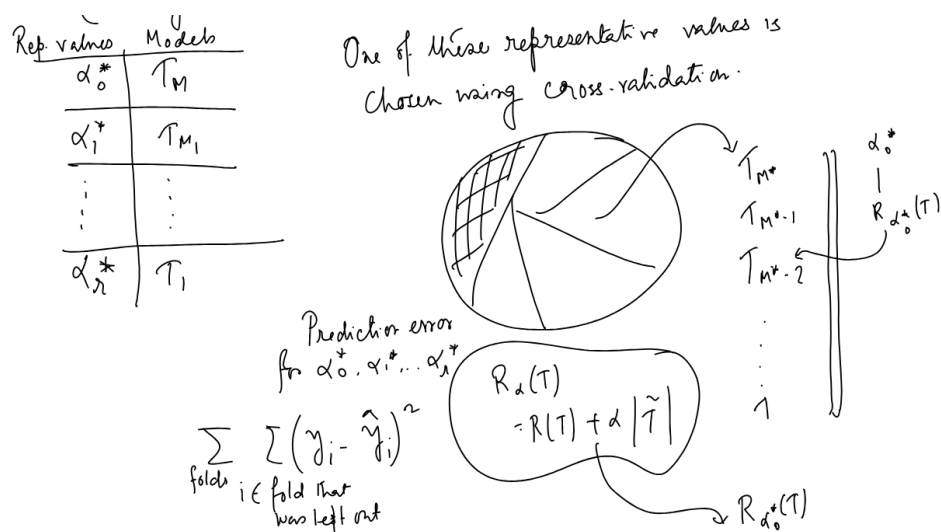
$$R(T_M) + \alpha M > R(T_k) + \alpha k \quad \Rightarrow \quad \alpha > \frac{R(T_k) - R(T_M)}{M - k}.$$

$$\alpha_0 = \min_{1 \leq k < M} \frac{R(T_k) - R(T_M)}{M - k}.$$

This process automatically deletes some of the candidates.

For each of these intervals, we choose a representative value (usually AM/GM of the upper & lower limits).

One of these representative values is chosen using cross-validation.



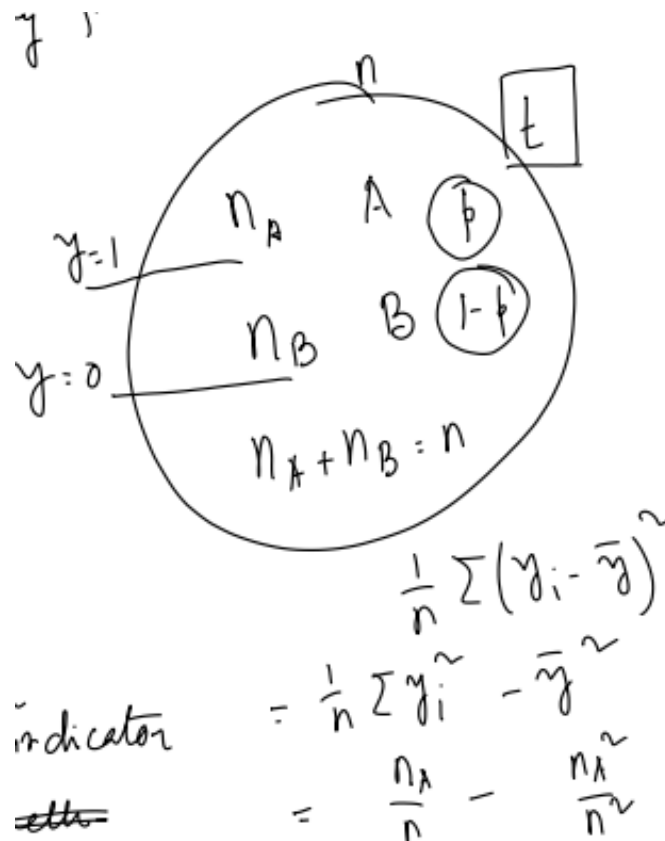
Prediction error for $\alpha_0^*, \alpha_1^*, \dots, \alpha_n^*$ is estimated by

$$\sum_{\text{folds } i \in \text{fold that was left out}} \sum (y_i - \hat{y}_i)^2$$

and the corresponding $R_{\alpha^*}(T)$ is selected.

Classification tree (Decision tree)

- We need to choose an appropriate impurity function.
- How to classify a future observation.



If we use the regression tree approach and use the variance as the impurity function, we get

$$i(t) = \frac{n_A n_B}{n^2} = p_A p_B = p(1-p).$$

(For binary classification, one can use an indicator variable and follow the regression tree algorithm for constructing the tree.)

A region corresponding to a leaf node t is assigned as a favourable region for class-1 or class-2 depending on whether $\bar{y}_t > 0.5$ or $\bar{y}_t \leq 0.5$.

What if there are more than two classes?

$$i(t) = \psi(p_1, p_2, \dots, p_K), \quad \sum_{i=1}^K p_i = 1.$$

Desirable properties

1. ψ should be symmetric in its arguments:

$$\psi(p_1, \dots, p_K) = \psi(p_{\pi(1)}, \dots, p_{\pi(K)})$$

for any permutation π of $\{1, \dots, K\}$.

2. ψ should be maximized when $p_i = 1/K$ for $i = 1, 2, \dots, K$.
3. ψ should be minimized when $p_i = 1$ for one i and $p_j = 0$ for the rest.

... ..

$p_i = 1$ for one, and $p_i = 0$ for the rest.

split

value of i and

$\sum_{i=1}^K p_i^* = 1$

$\sum p_i^* \leq \sum p_i = 1$

4. ψ should be such that the reduction in impurity for any split is non-negative.

If $\psi(p_1, \dots, p_K)$ is symmetric in its arguments and it is concave, it will satisfy all four properties mentioned above.

End of notes for W05

Statistical Learning – Week 06

Impurity Measures, Bagging, Random Forest & MARS

August 28, 2026

1 Properties of Impurity Function Ψ

- (i) symmetric in its arguments
- (ii) Ψ is max when $p_1 = \dots = p_k = 1/k$.

We shall show that for any $(p_1, \dots, p_k)$ with $\sum_{i=1}^k p_i = 1$,

$$\Psi(p_1, \dots, p_k) \leq \Psi(1/k, 1/k, \dots, 1/k)$$

$$\begin{aligned} \Psi(p_1, p_2, \dots, p_k) &= \Psi(p_2, \dots, p_k, p_1) = \Psi(p_3, \dots, p_k, p_1, p_1, p_2) \\ &= \Psi(p_k, p_1, p_2, \dots, p_{k-1}) = \Psi_0, \text{ say} \end{aligned}$$

$$\frac{1}{k} \left[(p_1, p_2, \dots, p_k) + (p_2, p_3, \dots, p_k, p_1) + \dots + (p_k, p_1, \dots, p_{k-1}) \right] = \left(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k} \right)$$

$$\begin{aligned} \Psi\left(\frac{1}{k}, \dots, \frac{1}{k}\right) &= \Psi\left(\frac{1}{k}(\alpha_1 + \dots + \alpha_k)\right) \\ &\geq \frac{1}{k}\Psi(\alpha_1) + \frac{1}{k}\Psi(\alpha_2) + \dots + \frac{1}{k}\Psi(\alpha_k) \\ &= \frac{1}{k}\Psi_0 + \frac{1}{k}\Psi_0 + \dots + \frac{1}{k}\Psi_0 = \Psi_0 = \Psi(p_1, \dots, p_k) \end{aligned}$$

- (iii) Ψ is min when one of the p_i 's is 1 and the rest are 0.

$$\Psi(1, 0, 0, \dots, 0) = \Psi(0, 1, 0, \dots, 0) = \dots = \Psi(0, 0, 0, \dots, 1) = \Psi^*, \text{ say}$$

We need to show that for any $(p_1, \dots, p_k)$ with $\sum_{i=1}^k p_i = 1$,

$$\Psi(p_1, p_2, \dots, p_k) \geq \Psi^*$$

$$(p_1, p_2, \dots, p_k) = p_1(1, 0, \dots, 0) + p_2(0, 1, 0, \dots, 0) + \dots + p_k(0, 0, \dots, 0, 1)$$

Using the concavity, we have

$$\begin{aligned} \Psi(p_1, p_2, \dots, p_k) &\geq p_1\Psi(1, 0, \dots, 0) + p_2\Psi(0, 1, 0, \dots, 0) \\ &\quad + \dots + p_k\Psi(0, 0, \dots, 1) \\ &= p_1\Psi^* + p_2\Psi^* + \dots + p_k\Psi^* = \Psi^* \end{aligned}$$

(iv) For any split of t , $\Delta i(t) \geq 0$.

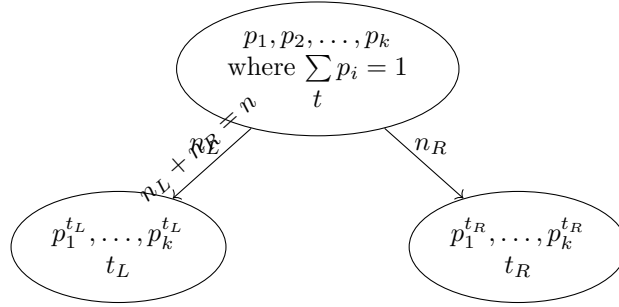


Figure 1: Impurity decrease for a split of node t into t_L and t_R

$$\begin{aligned}
 i(t) &= \Psi(p_1, \dots, p_k) \\
 i(t_L) &= \Psi(p_1^{t_L}, p_2^{t_L}, \dots, p_k^{t_L}) \\
 i(t_R) &= \Psi(p_1^{t_R}, p_2^{t_R}, \dots, p_k^{t_R}) \\
 \Delta i(t) &= \Psi(p_1, \dots, p_k) - \left\{ \frac{n_L}{n} \Psi(p_1^{t_L}, \dots, p_k^{t_L}) + \frac{n_R}{n} \Psi(p_1^{t_R}, \dots, p_k^{t_R}) \right\}
 \end{aligned}$$

We need to show that $\Delta i(t) \geq 0$ for any split.

Note that

$$\begin{aligned}
 p_i &= \frac{n_L}{n} p_i^{t_L} + \frac{n_R}{n} p_i^{t_R} \quad \text{for all } i = 1, 2, \dots, k. \\
 \underline{p} &= \frac{n_L}{n} \underline{p}_L + \frac{n_R}{n} \underline{p}_R \quad \text{where } \frac{n_L}{n} + \frac{n_R}{n} = 1 \\
 &\Rightarrow \Psi(\underline{p}) \geq \frac{n_L}{n} \Psi(\underline{p}_L) + \frac{n_R}{n} \Psi(\underline{p}_R) \\
 &\Rightarrow \Psi(\underline{p}) - \left[\frac{n_L}{n} \Psi(\underline{p}_L) + \frac{n_R}{n} \Psi(\underline{p}_R) \right] \geq 0 \\
 &\quad \underbrace{\Delta i(t)}_{\geq 0}
 \end{aligned}$$

2 Impurity of a Node – Definition and Popular Choices

Any concave function $\Psi(p_1, \dots, p_k)$ which is symmetric in its arguments, (max at uniform, min at vertices) can be used to define the impurity of a node.

Some popular choices are:

Gini Index

i) $\Psi(p_1, p_2, \dots, p_k) = \sum_{i \neq j} p_i p_j$ (This index $\sum_{i \neq j} p_i p_j$ is known as the Ginn's impurity index.

$$\begin{aligned}
 (p_1, p_2, \dots, p_k) &\times ?? \\
 \text{Gini index } \sum_{i \neq j} p_i p_j &\text{ is known as Gini's impurity index.}
 \end{aligned}$$

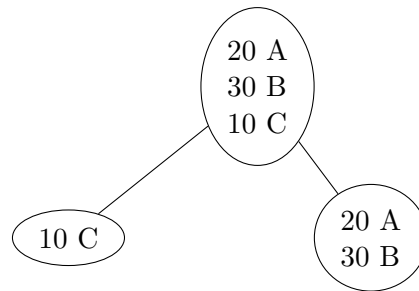


Figure 2: Example: Parent node 20 A, 30 B, 10 C split into pure 10 C node and mixed 20 A 30 B node. $\Psi(p_1, p_2, p_3) = 0$ would indicate purity, but here we have a mixture of observations from two classes.

$$\left(\sum p_i\right)^2 = 1$$

$$\sum p_i^2 + \sum_{i \neq j} p_i p_j = 1 \Rightarrow \sum_{i \neq j} p_i p_j = 1 - \sum_{i=1}^k p_i^2$$

Entropy

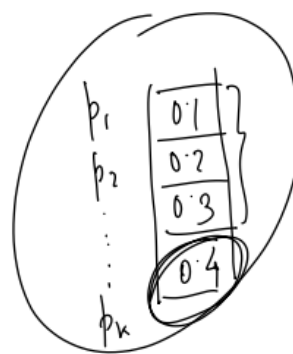
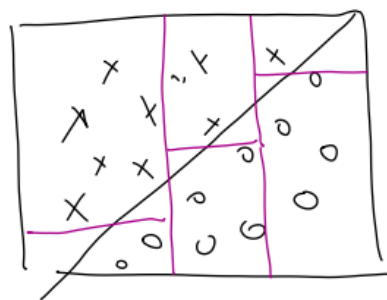
ii) Entropy

$$\Psi(p_1, p_2, \dots, p_k) = - \sum_{i=1}^k p_i \log p_i$$

Misclassification Impurity

iii) Misclassification impurity

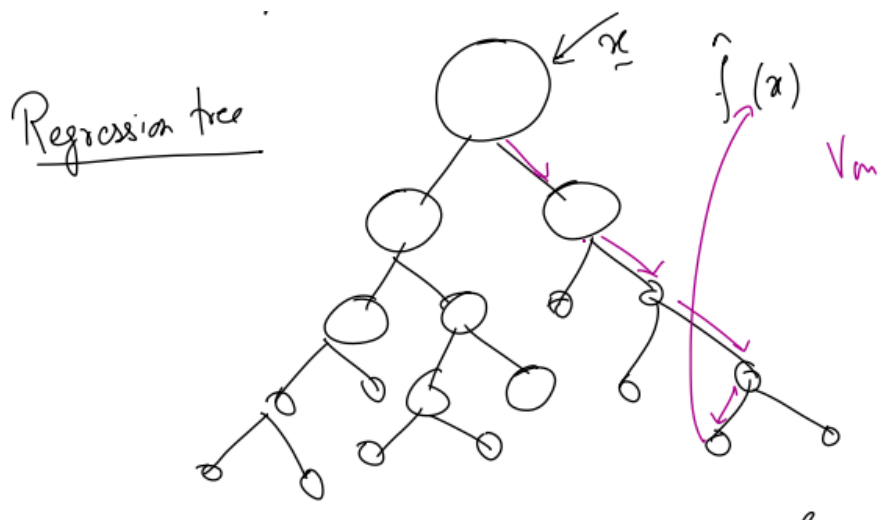
$$\Psi(p_1, \dots, p_k) = 1 - \max\{p_1, \dots, p_k\}$$



3 To Reduce Variability – Motivation for Ensembles

To reduce the variability of the regression tree / classification tree methods, often we construct several trees and aggregate the results.

Regression tree: $\hat{f}(x)$



$Var(\hat{f}(x))$ is high

$$Z_1, Z_2, \dots, Z_m \quad Var(Z_i) = \sigma^2, \quad i = 1, 2, \dots, m$$

$$Var(\bar{Z}) = \frac{\sigma^2}{m} \quad (\text{if } Z_i\text{'s are indep})$$

In order to reduce the variance of the regression tree estimator, we can use the following algorithm.

Data: $\{X_1, X_2, \dots, X_n; y_1, y_2, \dots, y_n\}$

Step 1: Generate a bootstrap sample of size n (A sample of size n is drawn from the data using SRSWR) $\rightarrow$ Simple random sampling with replacement

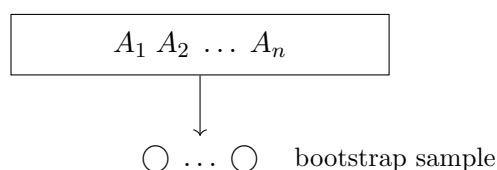


Figure 3: SRSWR - Simple random sampling with replacement

Expected no. of distinct objects in the sample?

Effective sample size:

$$N_0 = \sum_{i=1}^n \mathbf{1}_{\{A_i \text{ is in the sample}\}}$$

$$\begin{array}{l}
 \swarrow \quad \nwarrow \\
 1 \text{ w.p. } 1 - \left(\frac{n-1}{n}\right)^n \\
 0 \text{ w.p. } \left(\frac{n-1}{n}\right)^n
 \end{array}$$

$$E(N_0) = \sum_{i=1}^n E[\mathbf{1}_{\{A_i \text{ is in the sample}\}}] = \sum_{i=1}^n \left\{ 1 - \left(\frac{n-1}{n}\right)^n \right\} = n \left\{ 1 - \left(\frac{n-1}{n}\right)^n \right\}$$

$$E\left(\frac{N_0}{n}\right) = 1 - \left(\frac{n-1}{n}\right)^n$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} E\left(\frac{N_0}{n}\right) &= \lim_{n \rightarrow \infty} 1 - \left(\frac{n-1}{n}\right)^n = \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{1}{n}\right)^n \\
 &= 1 - \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = 1 - e^{-1} \approx 0.632 \approx 2/3
 \end{aligned}$$

Let $\{(X_{1i}^{(b)}, Y_{1i}^{(b)}), \dots, (X_{ni}^{(b)}, Y_{ni}^{(b)})\}$ be the sub-sample

Step 2: Based on this sub-sample, grow a large sized tree as before (but do not use pruning).

Let $f^{(b)}$ be the regression tree estimator obtained from this tree.

Repeat Step 1 and Step 2, several times (B times) to get $f^{(b)}$ for $b = 1, 2, \dots, B$ and use $\hat{f}(x) = \sum_{b=1}^B f^{(b)}(x)/B$ as the final estimate for $f(x)$.

This bootstrap aggregation method is known as **Bagging**.

4 Variance of Averaged Estimators

How to choose p' ? How many sub samples? What modifications are needed for classification tree?

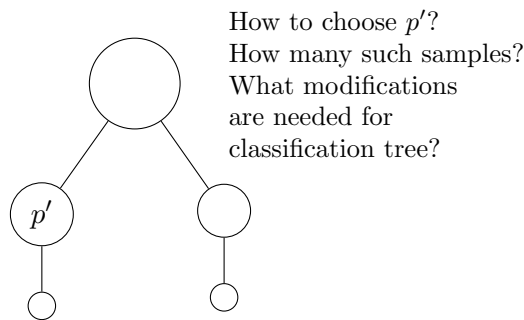
$$Z_1, \dots, Z_m \quad \text{Var}(Z_i) = \sigma^2$$

$$\bar{Z} = \frac{\sum Z_i}{m} \quad \begin{cases} E(Z_1) = \dots = E(Z_m) \\ \downarrow \\ E(\bar{Z}) = E(Z_1) = \dots = E(Z_m) \end{cases}$$

$\text{Var}(\bar{Z}) \approx \frac{\sigma^2}{m}$ if $Z_1, \dots, Z_m$ are independent

$$\begin{aligned}
 \text{Var}(\bar{Z}) &= \frac{1}{m^2} \left[\sum_{i=1}^m \text{Var}(Z_i) + \sum_{i \neq j} \text{Cov}(Z_i, Z_j) \right] \\
 &= \frac{1}{m^2} [m\sigma^2 + m(m-1)\rho\sigma^2]
 \end{aligned}$$

where $\rho = \text{Corr}(Z_i, Z_j)$



5 Aggregation of Trees

Aggregation of trees
 - Bagged regression trees
 - Random forest

5.1 Bagged Regression Tree

1. Generate a sub sample of size n using SRSWR.
2. Construct a regression tree based on that sub sample. (Use the same stopping criterion as before but do not use pruning)

Repeat Steps 1 and 2 several times (B many times)

Let $\hat{f}_1, \hat{f}_2, \dots, \hat{f}_B$ be the B regression tree estimators in these B cases. Then finally we use:

$$\hat{f}_{\text{BAG}}(x) = \frac{1}{B} \left(\hat{f}_1(x) + \dots + \hat{f}_B(x) \right) \text{ as the Bagged regression tree estimator.}$$

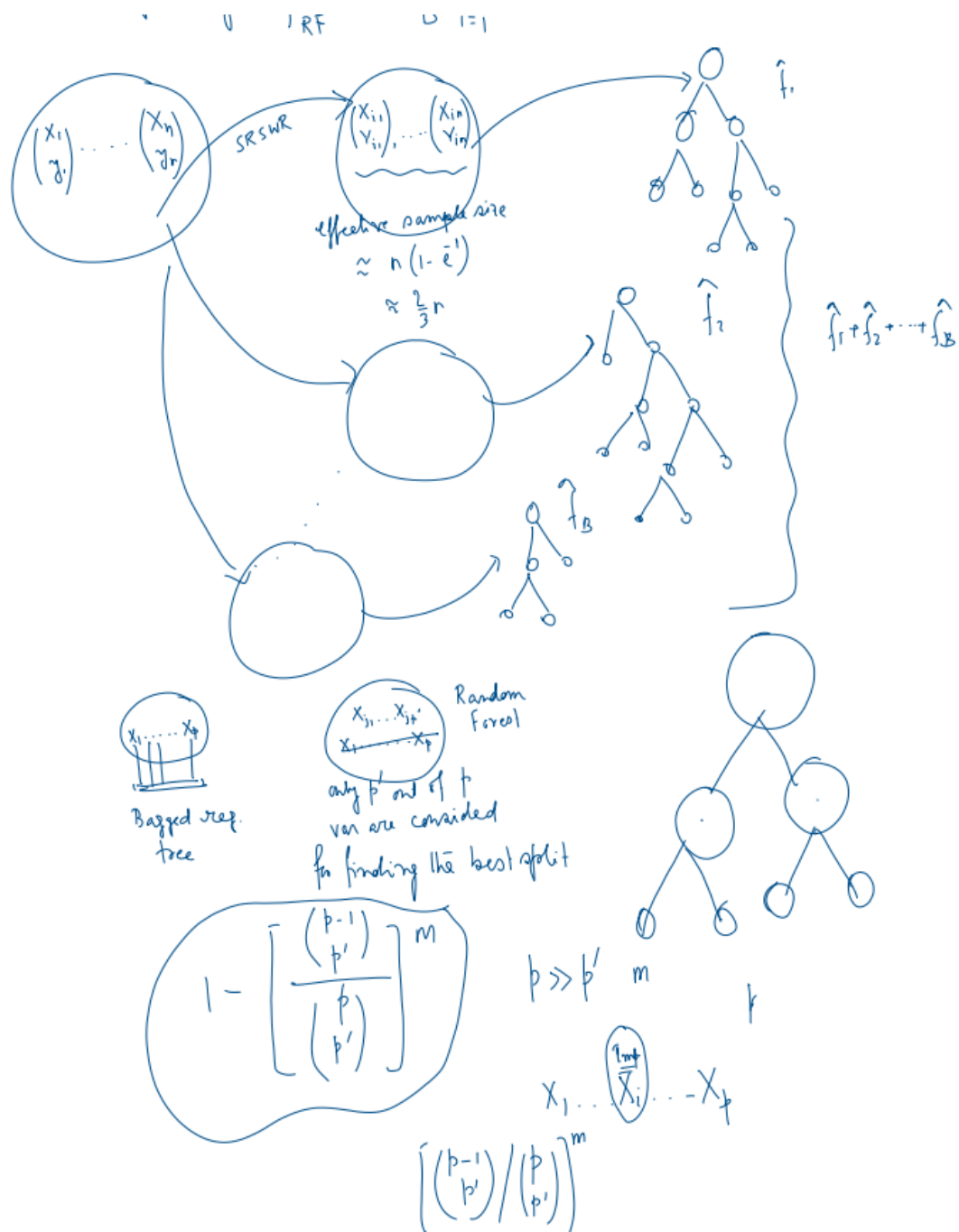
5.2 Random Forest

1. Generate a sub sample of size n using SRSWR.
2. Construct a regression tree based on that sub sample. The same stopping criterion can be used and prior pruning should be avoided. But unlike bagged regression trees, we do not consider all features, we consider only a subset of features at a time to find out the best split (chosen randomly) at a time to find out the best split.

Steps 1 and 2 are repeated several times.

If $\hat{f}_1^*, \hat{f}_2^*, \dots, \hat{f}_B^*$ are the regression tree estimators, then the random forest estimator of the regression function is given by:

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{i=1}^B \hat{f}_i^*(x).$$

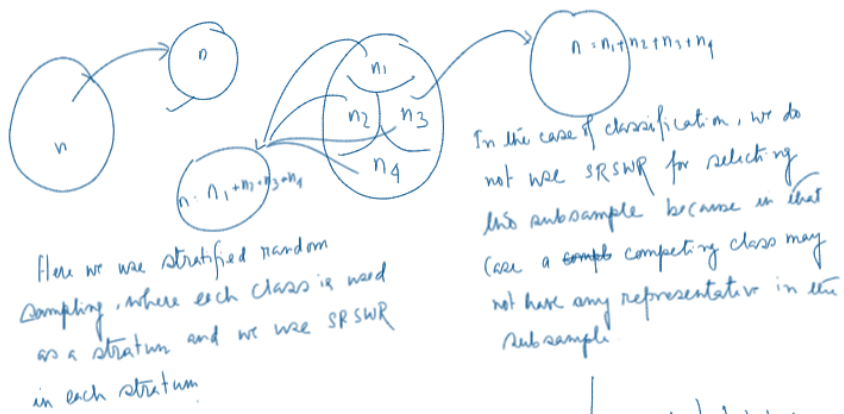
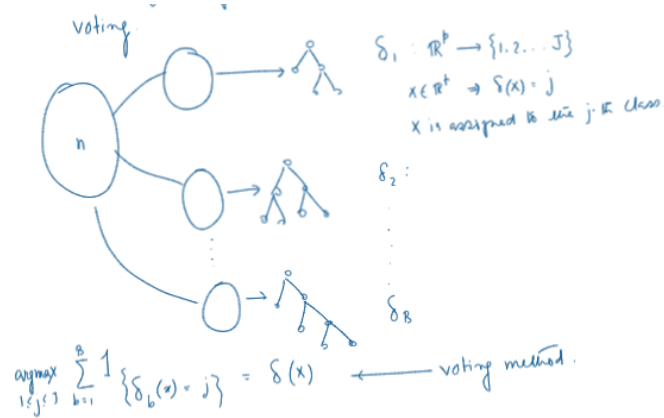


For classification, the procedure remains almost the same. (Sub sample selection & construction of trees are done in the same way) But instead of taking the average at the final step, we use voting.

$$S_i : \mathbb{R}^p \rightarrow \{1, 2, \dots, J\}$$

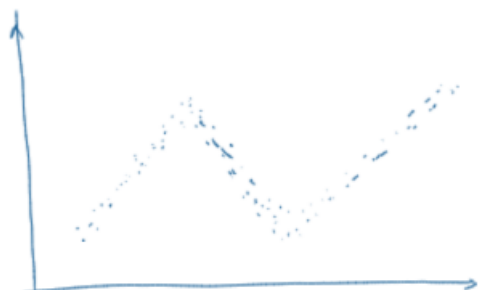
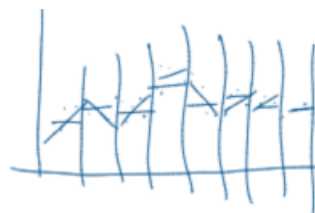
$$x \in \mathbb{R}^p \Rightarrow S(x) = j$$

x is assigned to the j -th class.



in each iteration:

MARS (Multivariable Adaptive Regression Splines)

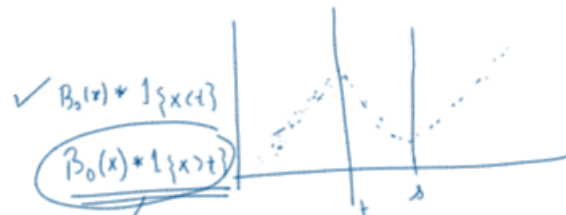
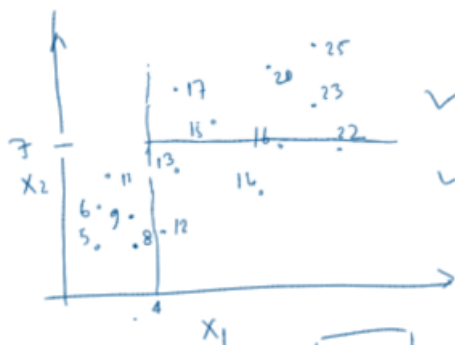


$$B_0(x) = 1$$

$$\sum (y_i - a B_0(x_i))^2$$

$$\sum (y_i - a)^2 \rightarrow a = \bar{y}$$

Regression tree



$$B_0(x) * 1\{x < t\}$$

$$B_0(x) * 1\{x > t\}$$



$$B_0(x) = 1$$

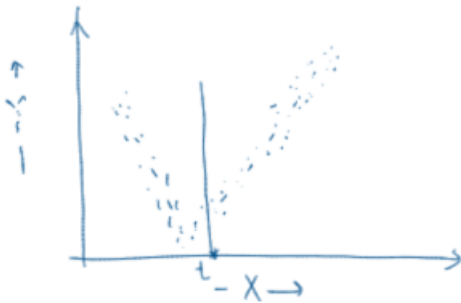
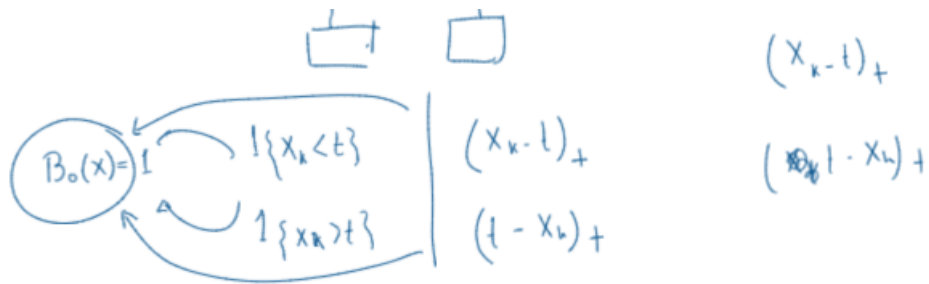
$$B_0(x) * 1\{x_1 > 4\}$$

$$B_0(x) * 1\{x_1 < 4\}$$

$$B_0(x) * 1\{x_1 > 4\} * 1\{x_2 > 7\}$$

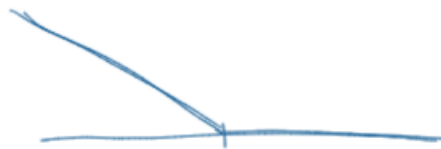
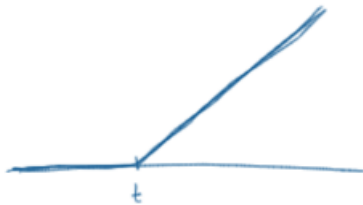
$$B_0(x) * 1\{x_1 > 4\} * 1\{x_2 < 7\}$$

$$(X_k - t)_+$$

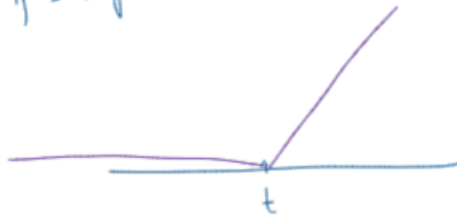


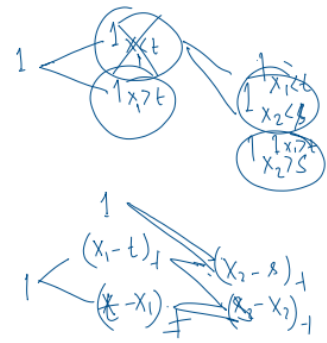
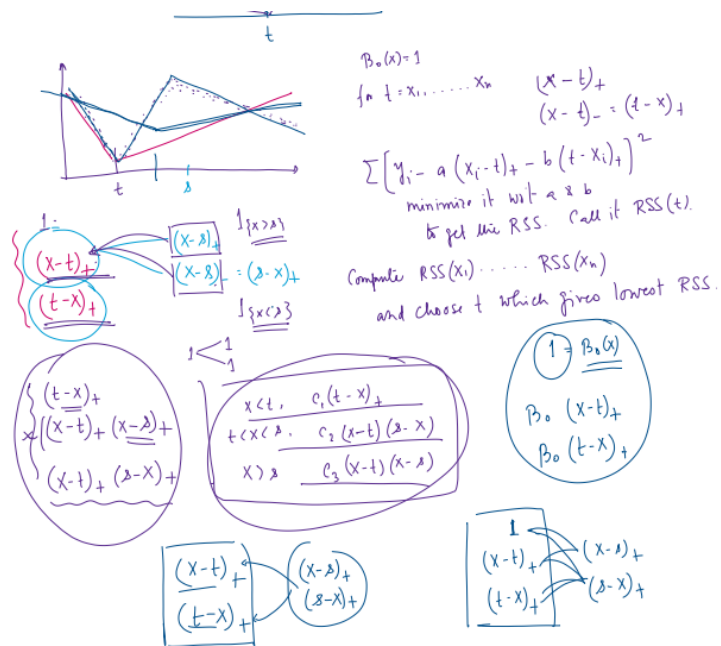
$B_0(x) = 1$
 Reg. based on this gives $\hat{f}(x) = \bar{y}$ for all x .

$$B_0(x) \begin{cases} B_0(x)(x-t)_+ = \begin{cases} x-t & \text{if } x > t \\ 0 & \text{o.w.} \end{cases} \\ B_0(x)(t-x)_+ = \begin{cases} t-x & \text{if } t > x \\ 0 & \text{o.w.} \end{cases} \end{cases}$$



What if I regress Y on these two basis functions?





Start with a basis $B_0 = 1$
 Suppose that after the m th stage, you have basis
 functions $B_0, B_1, B_2, \dots, B_m$.

Choose the var X_i
 Choose a cent t
 $\{x_{i1}, x_{i2}, \dots, x_{in}\}$

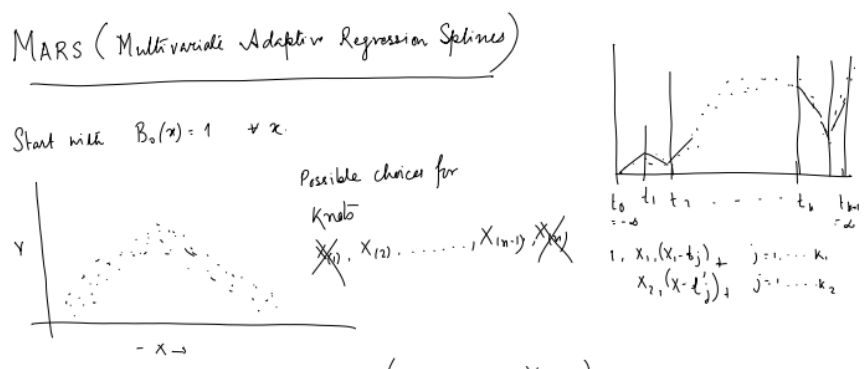
$B_0, B_1, \dots, B_m$
 $B_{m+1} = (x_i - t)_+ B_j$
 $B_{m+2} = (t - x_i)_+ B_j$
 $[j = 0, 1, \dots, m]$

$p \neq n \neq m$
 $(x_{i1} - t)_+, (x_{i2} - t)_+, \dots, (x_{in} - t)_+$

MARS (Multivariate Adaptive Regression Splines) and Classification / Bayes Decision Theory

Lecture Notes, Statistical Learning-I

1 MARS (Multivariate Adaptive Regression Splines)



Start with $B_0(x) = 1$ & x .

1.1 Possible choices for Knots

$$X_{(1)}, X_{(2)}, \dots, X_{(m-1)}, X_{(m)}$$

For each choice of the knot t ($t = X_{(1)}, \dots, X_{(m-1)}$) we consider a pair of functions

$$(x - t)_+ = \begin{cases} x - t & \text{if } x > t \\ 0 & \text{otherwise} \end{cases}$$

and

$$(t - x)_+ = \begin{cases} t - x & \text{if } t > x \\ 0 & \text{otherwise} \end{cases}$$

For this pair, we consider two new functions $B_0(x)(x - t)_+$ and $B_0(x)(t - x)_+$ and regress Y on $B_0(x)$, $B_0(x)(x - t)_+$, $B_0(x)(t - x)_+$.

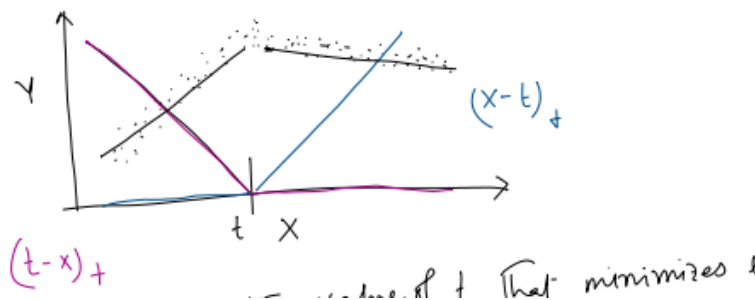
We minimize

$$\sum \left[Y_i - a_0 B_0(x_i) - a_1 B_0(x_i)(x_i - t)_+ - a_2 B_0(x_i)(t - x_i)_+ \right]^2$$

w.r.t. a_0, a_1 and a_2 and compute the corresponding RSS.

Finally, we shall choose the value of t that minimizes the residual SS. For instance if $t = X_{(k)}$ then we take $B_1(x) = B_0(x)(x - X_{(k)})_+$ & $B_2(x) = B_0(x)(X_{(k)} - x)_+$

$$B_0(x) \longrightarrow \{B_0(x), B_1(x), B_2(x)\}$$



2 Forward Step Continuation & Multivariate Case

Suppose that at some stage, we have m many basis functions

$$B_0(x), \dots, B_m(x).$$

At the next stage for all possible choices of t , we consider the pair $(x-t)_+$ & $(t-x)_+$ as before

$$\begin{array}{ll} B_0(x) & \\ B_1(x) & (x-t)_+ \\ \vdots & (t-x)_+ \\ B_m(x) & \end{array}$$

For each $j = 0, 1, \dots, m$ we consider two new functions $B_j(x)(x-t)_+$ and $B_j(x)(t-x)_+$ and compute the RSS for a model with $(m+1) + 2$ many basis functions

$$\{B_0, \dots, B_m, B_s(x)(t-x)_+, B_s(x)(x-t)_+\}$$

We do this for all choices of s and t and finally consider the one leading to the lowest RSS.

We proceed in this way and stop when the reduction in RSS is not significant.

2.1 Multivariate Case

For the multivariate case, the procedure remains almost the same.

There we start with $B_0(x) \sim 1$ as before.

At the next step, we have several choices for the pair:

$$\left. \begin{array}{l} (x_j - x_{(i)})_+ \\ (x_{(i)} - x_j)_+ \end{array} \right\} \begin{array}{l} \text{for } i = 2, \dots, n-1 \\ j = 1, 2, \dots, p \end{array}$$

For each of these pairs, we compute the RSS by minimizing

$$\sum \left[Y_n - a_0 B_0(x_n) - a_1 B_0(x_j)(x_{jn} - x_{j(i)})_+ - a_2 B_0(x_j)(x_{j(i)} - x_{jn})_+ \right]^2$$

w.r.t. a_0, a_1 and a_2 .

Finally we choose the pair leading to the lowest RSS.

Also, when we have $m+1$ many elements in the basis, we consider

$$\left. \begin{array}{l} B_s(x)(x_j - x_{j(i)})_+ \\ B_s(x)(x_{j(i)} - x_j)_+ \end{array} \right\}$$

as the possible pair of functions and compute the RSS for the model based on $(m+1) + 2$ many basis functions.

We do this for all choices of s ($s = 0, 1, \dots, m$), j ($j = 1, \dots, p$), and i ($i = 2, 3, \dots, n-1$) and finally select the one which gives the lowest RSS.

We proceed this way until the reduction in RSS after adding two new functions is negligible.

Interaction / Degree control

Sometimes, we put an upper bound on the order of interaction (or the degree of the polynomial).

$$\begin{array}{c}
 \text{1} \\
 \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \end{array} \right| \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \\ (x_1 - 7)_+ \\ (7 - x_1)_+ \end{array} \right| \left| \begin{array}{c} \text{1} \\ (x_1 - 3)_+ \\ (3 - x_1)_+ \\ (x_1 - 7)_+ \\ (7 - x_1)_+ \\ (x_1 - 3)_+ (x_3 - 11)_+ \\ (x_1 - 3)_+ (11 - x_3)_+ \end{array} \right|
 \end{array}$$

3 Backward Deletion & Classification / Regression Setup

After choosing the set of basis functions among many models we often use the backward deletion method for getting parsimonious models. During backward deletion, we get a nested sequence of models and one of them is finally chosen using an appropriate criterion.

Classification	Regression
$(X_1, y_1), \dots, (X_n, y_n)$	$(X_1, y_1), \dots, (X_n, y_n)$
$c_1, c_2, \dots, c_n \in \{1, 2, \dots, J\}$	$y_1, \dots, y_n \in \mathbb{R}$
A, B, ... T	real-valued
$\delta : \mathbb{R}^d \rightarrow \{1, 2, \dots, J\}$	Aim: $f : \mathbb{R}^d \rightarrow \mathbb{R}$
$x \in \mathbb{R}^d, \delta(x) \in \{1, \dots, J\}$	$x \in \mathbb{R}^d, f(x) \in \mathbb{R}$
class label	response value
$L(c, \delta)$: loss function	$L(y, f)$: loss function
$R(f) = E[L(c, \delta(x))]$	$Risk(f) = E[L(y, f(x))]$
e.g. $E[1_{\{c \neq \delta(x)\}}]$	e.g. $E(Y - f(x))^2$ sq. error loss
$= P(\delta(x) \neq c)$	$E Y - f(x) $ absolute error loss

3.1 Parametric Classifier (Two-class)

Consider a parametric classifier for a two-class problem

$$(x_1, 1), (x_2, 1), (x_3, 2), \dots, (x_n, 2)$$

$$c \in \{1, 2\}, \quad \delta \in \{1, 2\}$$

$$\delta(x) = \begin{cases} 1 & \text{if } \alpha + \beta x > 0 \\ 2 & \text{if } \alpha + \beta x < 0 \end{cases}$$

(consider a problem with one feature variable)

$$Risk = E[1_{\{c \neq \delta(x)\}}]$$

$$c \begin{cases} 1 & f_1 \leftarrow \text{density for class 1} \\ 2 & f_2 \leftarrow \text{density for class 2} \end{cases}$$

$$Risk(\delta) = P(\delta(x) \neq c) = P(c=1)P(\delta(x) \neq 1 | c=1) + P(c=2)P(\delta(x) \neq 2 | c=2)$$

3.2 Parametric Regression

Consider a parametric regression problem where we have only one predictor variable X and model we consider

$$f(x) = \alpha + \beta x$$

$$Risk = E[Y - f(x)]^2 = E[Y - \alpha - \beta x]^2$$

In practice, we minimize an empirical analog of the risk

$$\frac{1}{n} \sum (y_i - \alpha - \beta x_i)^2 \quad \text{or minimize} \quad \sum (y_i - \alpha - \beta x_i)^2$$

Handwritten notes and diagrams illustrating the risk function and its empirical version.

Risk function derivation:

$$\begin{aligned} Risk(\delta) &= P(\delta(x) \neq c) \\ &= P(c=1)P(\delta(x) \neq 1 | c=1) + P(c=2)P(\delta(x) \neq 2 | c=2) \\ &= \pi_1 \int_{\delta(x) \neq 1} f_1(x) dx + \pi_2 \int_{\delta(x) \neq 2} f_2(x) dx \end{aligned}$$

Bayesian rule for P(A):

$$P(A) = P(A \cap B) + P(A \cap B^c) = P(B)P(A|B) + P(B^c)P(A|B^c)$$

Empirical version of the risk:

$$\frac{1}{n} \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} + \frac{1}{n} \sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\}$$

Diagram: A Venn diagram showing two overlapping circles labeled n_1 and n_2 . The region n_1 is labeled "Class 1" and the region n_2 is labeled "Class 2". The regions are labeled $x_{11}, x_{12}, \dots, x_{1n_1}$ and $x_{21}, x_{22}, \dots, x_{2n_2}$.

4 Risk Expressions and Empirical Risk

$$= \pi_1 \int_{\delta(x) \neq 1} f_1(x) dx + \pi_2 \int_{\delta(x) \neq 2} f_2(x) dx$$

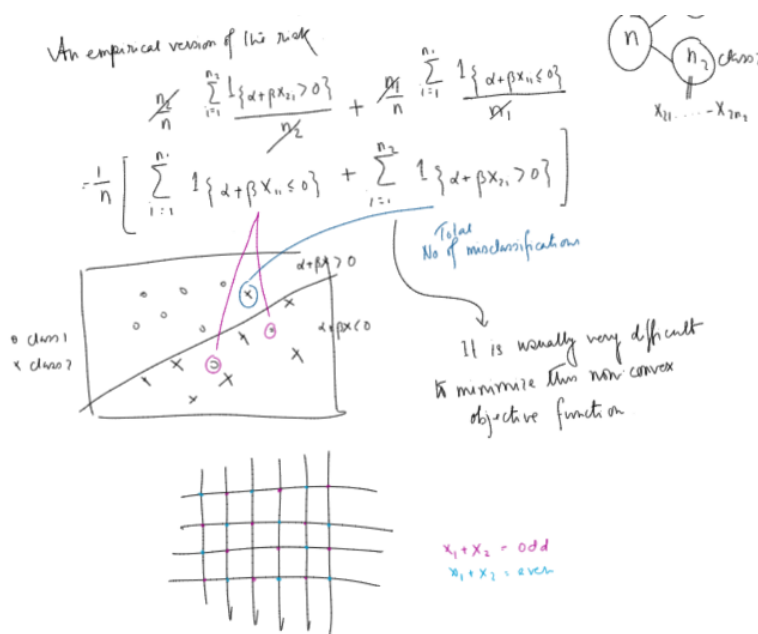
$$Risk(\delta) = P(\delta(x) \neq c) = P(\delta(x) = 1, c=2) + P(\delta(x) = 2, c=1) = P(\alpha + \beta x > 0, c=2) + P(\alpha + \beta x \leq 0, c=1)$$

An empirical version of the risk:

$$\frac{n_2}{n} \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} + \frac{n_1}{n} \sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\} = \frac{1}{n} \left[\sum_{i=1}^{n_1} 1\{\alpha + \beta x_{1i} \leq 0\} + \sum_{i=1}^{n_2} 1\{\alpha + \beta x_{2i} > 0\} \right]$$

Total number of misclassifications.

It is usually very difficult to minimize this non-convex objective function.



5 Two-Class Classification Risk (0-1 Loss)

In a two-class classification problem, if we use the 0-1 loss function, the risk (i.e. the expected loss) of a classifier δ is given by

$$P(\delta(X) \neq c).$$

If π_1 and π_2 are the prior probabilities of the two classes and f_1 and f_2 are their density functions, this can be expressed as

$$\begin{aligned} P(\delta(X) \neq c) &= P(\delta(X) = 2, c = 1) + P(\delta(X) = 1, c = 2) \\ &= P(c = 1)P(\delta(X) = 2 \mid c = 1) + P(c = 2)P(\delta(X) = 1 \mid c = 2) \\ &= \pi_1 \int_{x: \delta(x)=2} f_1(x) dx + \pi_2 \int_{x: \delta(x)=1} f_2(x) dx. \end{aligned}$$

An empirical version of this risk is given by

$$\frac{n_1}{n} \cdot \frac{1}{n_1} \sum_{i=1}^{n_1} I\{\delta(x_{i1}) = 2\} + \frac{n_2}{n} \cdot \frac{1}{n_2} \sum_{i=1}^{n_2} I\{\delta(x_{i2}) = 1\} = \frac{1}{n} \left[\sum_{i=1}^{n_1} I\{\delta(x_{i1}) = 2\} + \sum_{i=1}^{n_2} I\{\delta(x_{i2}) = 1\} \right],$$

the total proportion of misclassification.

Even for a simple classification method (eg linear classification) it may not be computationally feasible to minimize the empirical version of the risk over a collection of classifiers in that class. This is mainly because this empirical version is not a convex function.

Here one can think of minimizing

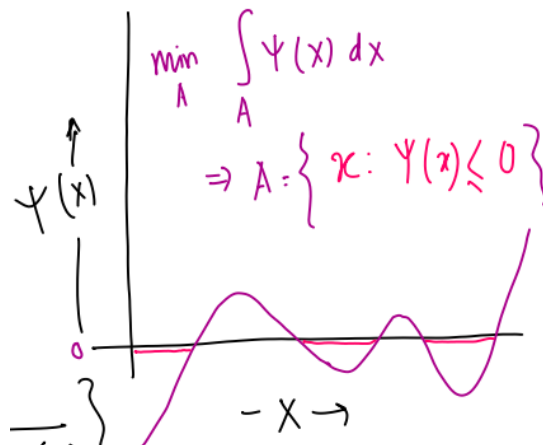
$$\begin{aligned} &\pi_1 \int_{x: \delta(x)=2} f_1(x) dx + \pi_2 \int_{x: \delta(x)=1} f_2(x) dx \quad \text{over } \delta. \\ &= \pi_1 \int_{R_2(\delta)} f_1(x) dx + \pi_2 \int_{R_1(\delta)} f_2(x) dx = \pi_1 \left\{ 1 - \int_{R_1(\delta)} f_1(x) dx \right\} + \pi_2 \int_{R_1(\delta)} f_2(x) dx \end{aligned}$$

$$= \pi_1 + \int_{R_1(\delta)} [\pi_2 f_2(x) - \pi_1 f_1(x)] dx.$$

This risk is minimized when

$$R_1(\delta) = \{x : \pi_1 f_1(x) > \pi_2 f_2(x)\} = \left\{x : \frac{\pi_1 f_1(x)}{\pi_1 f_1(x) + \pi_2 f_2(x)} > \frac{\pi_2 f_2(x)}{\pi_1 f_1(x) + \pi_2 f_2(x)}\right\} = \{x : p(1 | x) > p(2 | x)\}$$

(posterior probability of class 1).



Example, Defective Items

Company $\rightarrow$ Machine I $\rightarrow$ Machine II

- Machine I manufactures 60% of its products.
- 20% of the items produced by Machine I are defective.
- 30% of the items produced by Machine II are defective.

Purchased a pack of 20 items (5 defective, 15 non-defective) all handled by one machine.

Which machine is more likely to be the manufacturer?

6 Bayes Classifier – Machine Example & Lifetime Example

6.1 Defective Pack Example

Let X = number of defectives in a pack of 20.

$$P(M_1) = 0.6, \quad P(M_2) = 0.4 \quad (\pi_1, \pi_2)$$

$$X | M_1 \sim \text{Bin}(20, 0.2), \quad f_1(x) = \binom{20}{x} (0.2)^x (0.8)^{20-x}$$

$$X | M_2 \sim \text{Bin}(20, 0.3), \quad f_2(x) = \binom{20}{x} (0.3)^x (0.7)^{20-x}$$

$$P(M_1 | X = 5) = \frac{P(M_1)P(X = 5 | M_1)}{P(X = 5)} = \frac{\pi_1 f_1(5)}{\pi_1 f_1(5) + \pi_2 f_2(5)}$$

We assign the pack (here class $X = 5$) to the first class (M_1) if

$$\pi_1 f_1(5) > \pi_2 f_2(5).$$

Had x been the number of defectives, it would have been classified to M_1 if

$$\pi_1 f_1(x) > \pi_2 f_2(x).$$

Bayes classifier / Optimal classifier / Oracle classifier

$$\delta_{\text{Bayes}}(x) = \arg \max_{i=1,2} \pi_i f_i(x)$$

6.2 Lifetime Bulbs Example

- Company: 70% Type A – lifetime follows Exp with mean 2000 hrs.
- 30% Type B – lifetime follows Exp with mean 3000 hrs.

Purchased two bulbs of the same type. Observed their life times: 2500 hrs & 2700 hrs respectively.

$X = (X_1, X_2)$: lifetimes of two bulbs (feature vector)

$$f_1(x_1, x_2) = \frac{1}{2000} e^{-x_1/2000} \cdot \frac{1}{2000} e^{-x_2/2000} = \left(\frac{1}{2000}\right)^2 e^{-\frac{1}{2000}(x_1+x_2)}$$

$$f_2(x_1, x_2) = \left(\frac{1}{3000}\right)^2 e^{-\frac{1}{3000}(x_1+x_2)}$$

$$\pi_1 = 0.7, \quad \pi_2 = 0.3$$

$$\pi_1 f_1(x_1, x_2) > \pi_2 f_2(x_1, x_2)$$

leads to a linear decision boundary in the $(x_1 + x_2)$ plane.

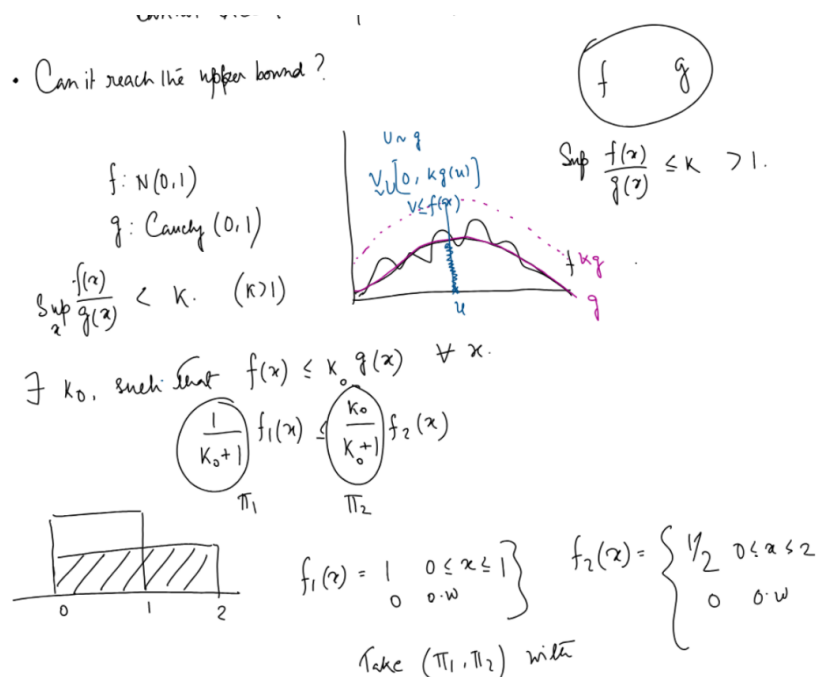
Bayes rule:

$$\delta(x) = \arg \max_i (\pi_i f_i(x)) = \arg \max_i p(i | x) = \arg \max_i \frac{\pi_i f_i(x)}{\sum_j \pi_j f_j(x)}$$

In a two-class classification problem, the Bayes risk (i.e. risk of the Bayes classifier) cannot exceed $\min\{\pi_1, \pi_2\}$. Can it reach the upper bound?

7 Properties of the Bayes Rule

7.1 Boundedness of Density Ratios



$$q \sim \text{Cauchy}(0,1)$$

Suppose $\frac{f(x)}{q(x)} < K$ ($K > 1$). There exists K_0 such that $f(x) \leq K_0 q(x) \quad \forall x$. Then

$$\frac{1}{K_0+1} f_1(x) \leq \frac{K_0}{K_0+1} f_2(x)$$

(with appropriate priors π_1, π_2).

7.2 Example with Uniform Densities

$$f_1(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad f_2(x) = \begin{cases} 1/2 & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Take (π_1, π_2) with $\pi_2 > 2/3$ (i.e. $\pi_1 < 1/3$).

7.3 Is the Bayes Rule Unique?

$$\delta(x) = \begin{cases} 1 & \text{if } \pi_1 f_1(x) \geq \pi_2 f_2(x) \\ 2 & \text{if } \pi_1 f_1(x) < \pi_2 f_2(x) \end{cases}$$

With $\pi_1 = 1/3$, $\pi_2 = 2/3$:

Region	$\pi_1 f_1(x)$	$\pi_2 f_2(x)$
(0, 1)	1/3	1/3
(1, 2)	0	1/3

Possible rules:

$$\delta_1(x) = \begin{cases} 1 & x \leq 1 \\ 2 & x > 1 \end{cases}$$

$$\delta_2(x) = \begin{cases} 1 & x \leq 0.5 \\ 2 & x > 0.5 \end{cases}$$

$$\delta_3(x) = \begin{cases} 1 & x \leq 0.1 \\ 2 & x > 0.1 \end{cases}$$

$$P(0.5 < X < 1) =$$

$$P(\delta_1(X) \neq \delta_2(X)) > 0$$

Hence the Bayes classifier is **not unique**.

7.4 Gaussian Case (Equal Covariance)

$$f_1 \sim N(\mu_1, \Sigma), \quad f_2 \sim N(\mu_2, \Sigma)$$

$$\pi_1 f_1(x) > \pi_2 f_2(x) \iff \log \pi_1 - \frac{1}{2}(x - \mu_1)^\top \Sigma^{-1}(x - \mu_1) > \log \pi_2 - \frac{1}{2}(x - \mu_2)^\top \Sigma^{-1}(x - \mu_2)$$

Property

Show that when the priors are equal, i.e. $\pi_1 = \pi_2 = 1/2$, the risk of this classifier is a decreasing function of the Mahalanobis distance

$$\sqrt{(\mu_1 - \mu_2)^\top \Sigma^{-1}(\mu_1 - \mu_2)}.$$

End of notes